IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 18 July 2025, accepted 9 August 2025, date of publication 15 August 2025, date of current version 21 August 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3599147

TOPICAL REVIEW

A Review on Quantum Machine Learning in Applied Systems and Engineering

YING-YI HONG[1], (Senior Member, IEEE), AND DYLAN JOSH DOMINGO LOPEZ[1,2], (Member, IEEE)

[1]Department of Electrical Engineering, Chung Yuan Christian University, Taoyuan 320, Taiwan
[2]Department of Computer Technology, De La Salle University, Manila 1004, Philippines

Corresponding author: Ying-Yi Hong (yyhong@ee.cycu.edu.tw)

This work was supported in part by the Ministry of Science and Technology of Taiwan under Grant NSTC 112-2221-E-033-010-MY3 and NSTC 114-2218-E-033-001.

ABSTRACT Quantum Machine Learning (QML) has emerged as a promising frontier within artificial intelligence, offering enhanced data-driven modeling through quantum-augmented representation, optimization, and learning capabilities, particularly in applied systems and engineering. This review comprehensively analyzes QML applications across various fields, including healthcare, communications, energy, space technology, transportation, and finance. The studies are categorized by learning modalities, QML architectures, and domain-specific use cases, emphasizing hybrid quantum-classical models deployed on current quantum hardware and simulators. Key insights underscore QML's strengths in handling high-dimensional, low-sample, and time-sensitive tasks. The review addresses critical infrastructural, operational, and ethical considerations for real-world implementation. The goal is to inform system engineers, researchers, and practitioners about the evolving capabilities and challenges of applying QML in practical domains.

INDEX TERMS Quantum machine learning, quantum computing, hybrid quantum computing, systems engineering.

NOMENCLATURE

Abbreviations

AAV	Autonomous Aerial Vehicle.
AI	Artificial Intelligence.
ANN	Artificial Neural Network.
CNN	Convolutional Neural Network.
CNOT	Controlled-NOT.
COBYLA	Constrained Optimization BY Linear Approximation.
CPU	Central Processing Unit.
CV	continuous-variable.
ECG	Electrocardiogram.
FCN	Fully connected Network.
GAN	Generative Adversarial Network.
GenAI	Generative Artificial Intelligence.
GNN	Graph Neural Network.
GPU	Graphics Processing Unit.
GRU	Gated Recurrent Unit.
HHL	Harrow–Hassidim–Lloyd.
HQCNN	Hybrid Quantum Convolutional Neural Network.
HQNN	Hybrid Quantum Neural Network.
LLM	Large Language Model.
LSTM	Long Short-Term Memory.
MERA	Multi-scale Entanglement Renormalization Ansatz.
ML	Machine Learning.
MLOps	Machine Learning Operations.
MRI	Magnetic Resonance Imaging.
NISQ	Noisy Intermediate-Scale Quantum.
PQC	Parameterize Quantum Circuit.
QAE	Quantum Autoencoder.
QAOA	Quantum Approximate Optimization Algorithm.
QBM	Quantum Boltzmann Machine.

The associate editor coordinating the review of this manuscript and approving it for publication was Md. Abdur Razzaque.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

QCBM	Quantum Constrained Boltzmann Machine.
QCNN	Quantum Convolutional Neural Network.
QDL	Quantum Deep Learning.
QDRL	Quantum Deep Reinforcement Learning.
QFT	Quantum Fourier Transform.
QGAN	Quantum Generative Adversarial Network.
QGRU	Quantum Gated Recurrent Unit.
QIL	Quantum Imitation Learning.
QLSTM	Quantum Long Short-Term Memory.
QML	Quantum Machine Learning.
QMLOps	Quantum Machine Learning Operations.
QNN	Quantum Neural Network.
QRAM	Quantum Random Access Memory.
QRL	Quantum Reinforcement Learning.
QRNN	Quantum Recurrent Neural Network.
QSGAN	Quantum semi-supervised Generative Adversarial Network.
QSSL	Quantum semi-supervised learning.
QSVM	Quantum Support Vector Machine.
QUBO	Quadratic Unconstrained Binary Optimization.
ResQT	Responsible Quantum Technology.
RGB	Red Green Blue.
RL	Reinforcement Learning.
RNN	Recurrent Neural Network.
SVM	Support Vector Machine.
VQA	Variational Quantum Algorithm.
VQC	Variational Quantum Circuit.
VQE	Variational Quantum Eigensolver.

I. INTRODUCTION

The advent of quantum computing has introduced a transformative computational paradigm with significant implications across both theoretical and applied fields. A diverse array of quantum computing devices has emerged, including quantum annealers, simulators, emulators, analog systems, and universal quantum computers, each tailored to solve specific problems. These technologies offer promising advantages in quantum simulation, combinatorial optimization, secure communication, and machine learning [1], [2], [3].

Despite its potential, the practical integration of quantum computing into industrial and applied engineering contexts remains a substantial challenge. Currently, the field of quantum computing is in a Noisy Intermediate-Scale Quantum (NISQ) era, marked by quantum processors with limited qubit counts and high susceptibility to noise. These constraints hinder the scalability needed for real-world engineering applications, which frequently involve high-dimensional datasets and complex modeling tasks in domains such as computer vision, healthcare, energy systems, and cyber-physical security.

Recent advances in quantum information science have paved the way for the convergence of quantum computing and machine learning, giving rise to the emerging field of quantum machine learning (QML). Foundational work in this area has demonstrated the feasibility of quantum-enhanced learning and feature extraction through approaches such as variational quantum algorithms (VQAs) [4], quantum support vector machines (QSVMs) [5], and parameterized quantum circuits (PQCs) [6]. These approaches have laid the groundwork for quantum neural networks (QNNs) and deeper architectures, collectively forming the backbone of quantum deep learning [7].

Nonetheless, QML in the NISQ landscape continues to face substantial obstacles, notably in terms of algorithmic robustness, hardware constraints, and real-world deployability. This review critically examines the role of QML from an applied engineering standpoint, encompassing a broad set of themes: algorithmic paradigms and learning modalities, domain-specific applications, computational infrastructure, and system-level engineering considerations. In addition to technical aspects, the discussion extends to contemporary concerns in AI development, including interpretability, ethical use, and responsible innovation within quantum contexts.

II. METHODS AND MATERIALS

Developments, design practices, and resource utility of quantum computing in machine learning and deep learning are the prime points of discussion in this review. Specific aspects such as cross-domain applications, various QML implementations, hardware and software resource use, and ethical/responsible technology use.

A. DATA SOURCES

A systematic search and retrieval of publications related to QML in applied engineering domains was conducted using multiple reputable academic publishers, including IEEE Xplore, SpringerLink, ScienceDirect (Elsevier), arXiv, and Nature, as illustrated in Figure 1. Each publication was manually reviewed and categorized according to metadata fields such as learning modality, application domain, quantum and classical methodologies, development tools, hardware platforms, and research activity. The resulting structured dataset comprises 172 entries, with publication years spanning from 2008 to 2024. Notably, the domain-specific application literature was scoped from 2021 to 2024, as shown in Figure 2.

Figure 2 further illustrates the topical research trends from 2021 to 2024, highlighting that publications on Quantum Deep Learning (QDL) have experienced a steeper exponential growth compared to general QML studies. Additionally, a trend line for Responsible Quantum Technology (ResQT) has been included to reflect the growing body of ethical and governance-oriented research that supports the development of applied QML systems. Particular emphasis was placed on identifying studies that demonstrate QML implementations in real-world or near-practical engineering contexts. Priority was given to publications featuring simulations, hybrid quantum-classical architectures, or algorithmic proposals accompanied by detailed resource analyses such as

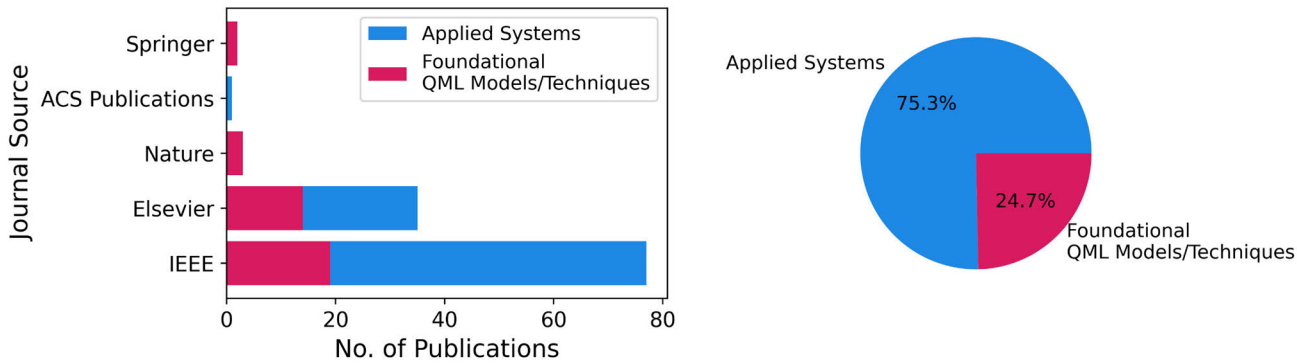


FIGURE 1. Summary of literature per journal publisher.

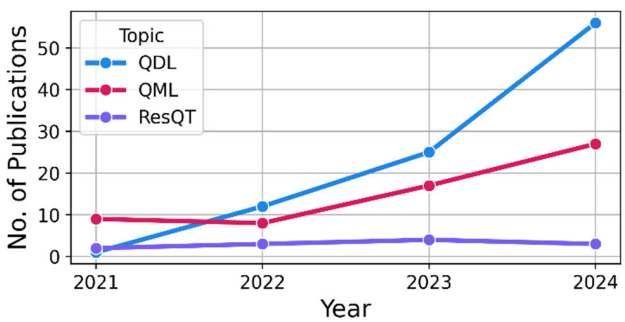


FIGURE 2. Year-based topic trend in this review.

qubit counts and circuit depth. Foundational and theoretical works were also included to provide context and support for the applied methodologies.

To minimize selection bias, sources were chosen to represent a diverse range of learning modalities, including Quantum Reinforcement Learning (QRL), Variational Quantum Circuits (VQC), Quantum Convolutional Neural Networks (QCNN), and Quantum Autoencoders (QAE) as well as various engineering sub-domains such as healthcare, energy systems, communications, and anomaly detection. Furthermore, aspects related to responsible AI and quantum technology, including explainability, ethical implications, and robustness, were incorporated by including publications indexed under terms such as "responsible quantum," "responsible artificial," and related keywords across the databases.

All publication links were verified to be accessible via institutional subscriptions or open-access channels at the time of writing. The curated dataset serves as the foundation for the subsequent analysis of learning models, domain-specific use cases, infrastructure requirements, and the principles of responsible QML design.

B. FILTRATION CRITERIA

To ensure the relevance, quality, and coherence of this review, a multi-stage filtration process was implemented. This process aimed at retaining publications that exhibit technical rigor and demonstrate the direct applicability of QML methods within engineering domains.

1) INCLUSION

Peer-review journals articles focused on QML foundational models, operations, and domain-specific applications were considered from the years 2021 to 2024 within the constraints of the NISQ computing era. Methodological specificity was also a key consideration, with preference given to papers that provided detailed implementations of QML models (e.g., VQAs, QNNs, QCNNs) along with algorithmic insights or performance benchmarks. Additional priority was assigned to studies that explicitly utilized real quantum hardware, cloud-based simulators, and established QML software platforms such as Qiskit, PennyLane, and Cirq.

2) EXCLUSION

Papers were excluded if they focused solely on theoretical or foundational aspects of quantum computing without demonstrating direct applications to machine learning or engineering practice. Studies that lacked sufficient technical detail, such as algorithmic descriptions or performance metrics, were omitted, as were duplicate publications or secondary reports that did not offer significant methodological advancements. Articles referencing quantum computing and machine learning independently, without a substantive integration of the two fields, were considered out of scope. Finally, papers for which the full text was unavailable or inaccessible at the time of review were excluded to ensure consistency and transparency in the analysis.

C. CATEGORIZATION

Other QML reviews, such as [8], [9], and [10], initially categorized prior work by algorithm similarity, such as QML and QDL distinctions—specifically, [8] and [10] categorized the papers by field of application. Further categorization of medical application cases was presented in the works of Chow [11], [12].

This review further categorizes the prior work into quantum machine learning, quantum deep learning, and application areas. Classification as quantum machine learning,

similar to classical ML, tackles a more generic scheme of data learning, including supervised learning, unsupervised learning, and reinforcement learning. A separate classification is dedicated to deep learning, given the span and volume of existing research. The QDL classification is further sub-categorized into various deep learning architectures similar to those in classical deep learning, with the addition of hybrid quantum deep learning, as in [8] and [10]. The last classification perspective is based on the technical and industrial application of prior research. This is sub-categorized based on top technical and engineering applications, as well as matching industry domains. Further categorization of application areas was inspired by specific classifications made in [12].

III. QUANTUM COMPUTING

The limitations of classical computing are increasingly constrained by hardware capabilities. In response, quantum computing has emerged as a promising alternative, gaining significant attention and development over the past decade. A thorough understanding of quantum mechanics, information theory, and computational principles is essential when designing and developing algorithms and applied systems based on QML.

A. QUANTUM INFORMATION

Data plays a critical role in all computing systems, particularly in the context of machine learning. In quantum systems, it is essential to understand the nature and representation of data. The fundamental unit of information in a quantum system is the quantum bit, or *qubit*. Unlike bits or classical bits that exist in states 0 or 1, a qubit can exist in a superposition of states simultaneously represented as $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, whereas a and β are complex probability amplitudes satisfying Born's rule $|\alpha|^2 + |\beta|^2 = 1$.

When multiple qubits are combined, they form the basis for quantum memory, a system capable of storing and preserving quantum states over time for subsequent retrieval, manipulation, or computation. Quantum memory enables access to classical data via architectures such as quantum random access memory (QRAM). The stored quantum states reside in a Hilbert space $\mathcal{H}$, where each memory element corresponds to a quantum vector $|\psi\rangle \in \mathcal{H}$. An n-qubit quantum system's Hilbert space can be defined as $\left(\mathbb{C}^2\right)^{\otimes n}$ which grows exponentially by a factor of 2^n. The exponential allows quantum systems to embed data in exponentially large feature spaces using linear resources.

A fundamental constraint in quantum information processing is the *no-cloning theorem*, which asserts that it is inherently impossible to create an identical copy of an arbitrary unknown quantum state. Unlike classical systems, where data can be freely duplicated, quantum systems prohibit such duplication due to the linearity and unitarity of quantum mechanics [13]. This limitation has significant implications for data security, imposes constraints on error correction techniques, and necessitates alternative data reuse and communication strategies within QML models. Despite these challenges, quantum memory capitalizes on the structure of Hilbert space by preserving superposition and entanglement over time, unlike classical RAM, which stores bits independently. This capability enables quantum systems to retain complex correlations and represent entire datasets as quantum states, thereby allowing for sublinear data access and exponential compression in machine learning tasks [14].

Another critical quantum resource is entanglement, a non-classical correlation in which the states of two or more qubits become interdependent, such that the state of one cannot be fully described without reference to the others, regardless of the spatial distance between them. This phenomenon, known as non-locality, enables quantum systems to model and propagate complex dependencies within data. Entanglement enhances computational expressiveness, facilitates parallelism, and underpins advanced tasks such as kernel evaluation, pattern recognition, and feature encoding in QML. However, it is important to emphasize that although entangled states exhibit instantaneous correlations, they do not permit faster-than-light communication, preserving causality following the principles of relativity.

B. QUANTUM ALGORITHMS

Quantum algorithms exploit the mathematical structure and physical behavior of quantum systems such as superposition, entanglement, and interference to solve problems more efficiently than classical counterparts. Several distinct classes of quantum algorithms have emerged; each tailored to different computational paradigms.

1) QUANTUM SIMULATION ALGORITHMS

Quantum simulation algorithms use quantum computers to model the evolution of physical systems, addressing problems that are classically intractable due to the exponential growth of quantum state spaces. These simulations are grounded in the time-dependent Schrödinger (1).

$$i\hbar\frac{\partial}{\partial t}\,|\psi\,(t)\rangle = H\,|\psi\,(t)\rangle \tag{1}$$

wherein (1) describes how quantum state $|\psi\,(t)\rangle$ evolves under a system's Hamiltonian H. Quantum algorithms implement this evolution via approximations such as Trotterization, Taylor expansion, or qubitization, enabling applications in chemistry, materials science, and molecular dynamics. A key example is Quantum Phase Estimation (QPE), used to extract eigenvalues of Hamiltonians, often in combination with hybrid algorithms like VQE. Another practical formulation frequently encountered in quantum computing is the Quadratic Unconstrained Binary Optimization (QUBO) problem. QUBO provides a mathematical framework for representing a wide range of combinatorial optimization tasks. It involves minimizing a quadratic objective function defined over binary variables as seen in (2).

$$\min_{x\in\{0,1\}^n} x^T Q x \tag{2}$$

where Q is a real-valued upper-triangular matrix and x is a binary vector with a length of n. QUBO models naturally map to the Ising Hamiltonian, making them well-suited for implementation on quantum annealers.

Analog simulation, using platforms like Rydberg atom arrays or trapped ions, allows real-time emulation of Hamiltonian dynamics without requiring discrete gate-based circuits. These systems are valuable for modeling many-body physics and benchmarking quantum behavior. Quantum simulation algorithms are applicable for optimization applications through quantum annealers, provided the problem is formulated as a QUBO model. Quantum optimization is generally applicable to operations research and is integrable with machine learning optimization routines.

2) ORACLE-BASED ALGORITHMS

Oracle-based quantum algorithms solve problems by querying a closed-box function, or oracle, that marks desirable solutions without revealing their structure. These algorithms are designed to reduce the number of queries needed to identify a solution compared to classical approaches. The most prominent example is Grover's algorithm, which provides a quadratic speedup for unstructured search problems. Given a function, $f : \{0, 1\}^n \rightarrow \{0, 1\}$, where the goal is to find an input x such that $f(x) = 1$. Grover's algorithm can find such x in $O\left(\sqrt{N}\right)$ queries, compared to $O(N)$ classically. Oracle circuits are also commonly used as benchmarking tools for quantum hardware due to their relatively shallow depth and theoretical guarantees on query complexity. However, implementing general-purpose oracles on real hardware remains a challenge, as each oracle must be custom-designed for the specific problem instance and circuit layout [15].

3) ALGEBRAIC ALGORITHMS

Algebraic quantum algorithms are a class of routines that exploit the algebraic structure of mathematical problems, particularly in number theory and group theory, to achieve exponential speedups over classical approaches. These algorithms typically rely on the Quantum Fourier Transform (QFT) and period-finding subroutines, which are central to solving problems like integer factorization, order finding, and discrete logarithms. Unlike oracle-based algorithms, which treat the problem as a closed-box and offer quadratic speedups via amplitude amplification, algebraic algorithms leverage known mathematical structure such as periodicity and symmetry to extract hidden parameters with greater efficiency using QFT-based routines.

The most popular manifestation is Shor's algorithm, which factors an integer n in polynomial time compared to the best-known classical algorithms that run in sub-exponential time. The core of Shor's algorithm involves finding the period of a function $f(x) = a^x \bmod N$, which is achieved through a quantum circuit that includes QFT to identify periodicity with high probability. Despite their power, algebraic algorithms typically require deep circuits and high qubit counts, making them less practical on current NISQ hardware.

4) VARIATIONAL QUANTUM ALGORITHMS

VQCs are a class of hybrid quantum-classical algorithms designed explicitly for near-term quantum hardware. These algorithms use PQCs, where gate operations depend on tunable parameters optimized by a classical routine to minimize or maximize a predefined objective function. VQCs are particularly suited to NISQ devices due to their shallow circuit depths and tolerance to limited coherence times.

The variational workflow begins by preparing an initial quantum state through a PQC, then measuring an observable (typically the expectation value of a Hamiltonian) and feeding that result into a classical optimizer. The optimizer updates the parameters to reduce the cost function, iterating this process until convergence. This hybrid architecture allows quantum processors to focus on state preparation and measurement while classical processors handle optimization and error resilience. Two of the most prominent variational algorithms are:

Variational Quantum Eigensolver. Used for estimating ground-state energies of molecular Hamiltonians, central to quantum chemistry applications. VQE approximates eigenstates by minimizing the expectation value $\left\langle \psi\left(\vec{\theta}\right) |H| \psi\left(\vec{\theta}\right)\right\rangle$, where $\vec{\theta}$ represents trainable parameters.

Quantum Approximate Optimization Algorithm Designed to solve combinatorial optimization problems by alternating between a cost Hamiltonian (which encodes the problem) and a mixing Hamiltonian. QAOA circuits are parameterized and optimized similarly to VQE but tailored for discrete optimization like QUBO.

While variational methods are highly adaptable, they come with challenges. Issues such as barren plateaus (regions of flat gradients), hardware noise, and optimizer instability can degrade performance, particularly as circuit width or depth increases. Nonetheless, VQCs are among the most promising approaches for demonstrating practical advantages on NISQ devices and are widely applied in simulation, optimization, and early-stage quantum learning architectures. An example of VQE utility lies in simulating Hamiltonians in quantum chemistry and similar material science studies. Combinatorial optimization problems, e.g., traveling salesman and knapsack problems, are formulated in QUBO models. Models are then optimized using VQA techniques via QAOA.

5) APPLICATION-SPECIFIC ALGORITHMS

Application-specific quantum algorithms are designed to solve well-structured problems within particular domains by combining quantum subroutines from previous methods like amplitude estimation, state preparation, and unitary evolution. Unlike general-purpose algorithms, these routines are tightly integrated with specific computational tasks such as solving linear systems, sampling distributions, or optimizing

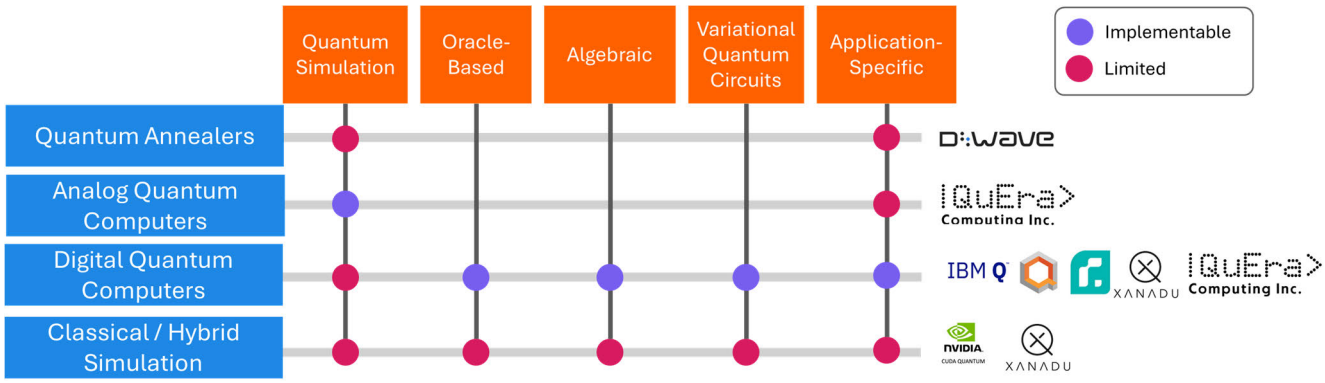


FIGURE 3. Applicability of quantum computing algorithms with different types of quantum computing devices.

domain-specific objective functions. A notable example is the HHL algorithm, which solves a linear system $Ax = b$.

Other examples include quantum amplitude estimation for accelerating Monte Carlo methods and quantum walks for solving routing and graph traversal problems. Variants of Grover's algorithm have also been adapted for specific use cases, such as clause satisfaction and collision detection. Although these algorithms often promise polynomial or exponential speedups in theory, their practicality depends on factors such as data encoding, problem sparsity, and hardware limitations. Oracle-based methods, such as Grover's algorithm, can be applied to unsorted database searches. Quantum amplitude estimation, on the other hand, is useful for estimating probability density functions or stochastic samples that may be present in deep learning. QFT can be applied in the Harrow–Hassidim–Lloyd (HHL) algorithm to solve linear systems of equations. Linear solvers can aid in machine learning applications, particularly with linear regression models.

C. QUANTUM COMPUTING DEVICES

The performance and scalability of QML models are closely tied to the capabilities and limitations of the underlying hardware. Various quantum computing paradigms have been developed, each optimized for different types of tasks. This section categorizes these devices into four primary types: quantum annealers, analog quantum computers, universal gate-based quantum computers, and classical quantum simulators. A summary of the type of quantum algorithms with applicable quantum hardware can be seen in Figure 3.

1) QUANTUM ANNEALERS

Quantum annealers are designed to solve binary or integer optimization, using adiabatic quantum computation, where the system evolves slowly from an initial ground state to the solution state of a problem Hamiltonian. Devices such as those developed by D'Wave are among the most accessible commercial implementations. They operate on a relatively large number of qubits (>5000 in recent models) but are limited in general-purpose programmability [16]. Qubit preparation in annealers typically involves initializing the system into a low-energy configuration and allowing quantum tunneling and entanglement to explore energy landscapes [17].

2) ANALOG QUANTUM COMPUTERS

Analog quantum devices simulate specific quantum systems by leveraging naturally occurring physical dynamics, such as interactions between neutral atoms, Rydberg excitations, or cold atoms in optical lattices [18], [19]. Analog systems do not apply discrete logic gates but instead evolve according to a continuously controlled Hamiltonian. A leading example is QuEra Computing's Aquila platform [20], which employs Rydberg atom arrays trapped by optical tweezers to model many-body quantum systems with tunable interaction graphs. Aquila supports up to 256 qubits and is highly effective for analog Hamiltonian simulation, especially in condensed matter and statistical physics problems at the time of this writing.

3) DIGITAL QUANTUM COMPUTERS

Universal quantum computers are also referred to as gate-based digital quantum computers. They operate by applying a series of quantum logic gates to qubits, similar to how classical computers use logical operations on bits. These systems are programmable, capable of simulating any unitary evolution, and form the backbone of modern quantum algorithms used in QML, optimization, and quantum simulation. Digital quantum computers prepare qubits in defined basis states, typically the ground state $|0\rangle$ and evolve these using unitary gate operations, such as Hadamard, CNOT, Pauli rotations, and parameterized rotation gates. These operations are composed of quantum circuits executed on physical qubits realized via different modalities. Several prominent commercial and academic implementations include superconducting quantum computers that may use superconducting transmon qubits controlled via microwave pulses [21] or emphasizing fast two-qubit gates and scalable lattice topologies [22]. Trapped-ion qubits can also be used where quantum gates are performed via precision laser pulses manipulating electronic

states of individual ions [23]. Photonic quantum computers employ light-based approach to digital quantum computing through continuous-variable (CV) photonic systems. Instead of qubits, CV platforms operate on modes of light [24].

Digital quantum computing is currently confined to the NISQ era such that quantum processors have limited qubit counts (typically <100), short coherence times, and no full-scale error correction. Computations are constrained to complete within microseconds since qubits are susceptible to the loss of quantum state due to environmental interactions or *decoherence*. Furthermore, gate noise and readout errors introduce inaccuracies in quantum circuits. Limited qubit connectivity, common in superconducting architectures like IBM and Rigetti, requires additional SWAP gates that increase circuit depth and compound noise [25]. These limitations constrain model complexity but also drive innovation in noise-resilient algorithms and efficient circuit design.

4) QUANTUM SIMULATIONS ON CLASSICAL HARDWARE (CPU/GPU)

In the current NISQ era, where physical quantum hardware is constrained by noise, limited qubit counts, and coherence times, classical quantum simulators play a crucial role in the development and testing of quantum algorithms. These simulators emulate quantum systems using conventional CPU or GPU architectures, enabling researchers to explore quantum circuit behavior, validate theoretical constructs, and benchmark performance under both idealized and noisy conditions.

Classical simulators most commonly use the state vector model, in which an n-qubit system is represented as a complex vector of size 2^n. Gate operations are implemented as unitary matrix multiplications, and measurements are simulated by probabilistic sampling of the resulting amplitudes. Several software platforms support CPU-based quantum simulations, including Qiskit Aer (IBM) [26], Cirq (Google) [27], QuTiP [28], Pennylane [29], ProjectQ [30], and CUDA Quantum [31]. These tools provide low-level access to quantum circuits, enable gate-by-gate debugging, and often include support for custom noise models, allowing detailed studies of algorithmic robustness in realistic settings.

To address computational bottlenecks, recent advancements in GPU-accelerated simulation have emerged. NVIDIA's cuQuantum SDK [32] offers optimized libraries such as cuStateVec for state-vector evolution and cuTensorNet for simulating tensor network-based circuits. These libraries leverage GPU parallelism to accelerate matrix operations and enable simulation of deeper circuits and larger systems compared to CPU-only tools.

IV. QUANTUM MACHINE LEARNING

Classical machine learning has progressed significantly through advances in pattern recognition and feature extraction. The integration of quantum computing and quantum information theory into machine learning commonly referred to as QML is expected to introduce improvements that surpass classical implementations. This stems from the ability of quantum systems to encode, manipulate, and extract patterns in high-dimensional Hilbert spaces that are inaccessible to classical systems [3], [33]. Quantum algorithms can exploit properties such as superposition, entanglement, and interference, which may enable exponential compression of data or novel learning paradigms. This section explores the design, capabilities, and implementation of QML models across different computational modalities. In addition to presenting models and frameworks, we begin by clarifying the three foundational components of quantum machine learning: data representation, feature extraction, and learning methods. Understanding these components is essential for analyzing how quantum models process information and how they differ from their classical counterparts.

A. DATA REPRESENTATION

In QML, classical data must be encoded into quantum states using methods known as quantum embeddings. The choice of encoding directly influences the model's expressivity and hardware compatibility. Quantum kernel methods, such as the QSVM, use feature map encodings that map classical inputs into high-dimensional Hilbert spaces for kernel-based learning [5]. In contrast, PQC models, including quantum neural networks, typically use Amplitude Encoding to load data into state amplitudes [34], Angle Encoding to map data onto gate rotation angles [6], or Phase Encoding to shift relative quantum phases [14]. Other advanced approaches include hybrid strategies that combine discrete and continuous variable encodings. For instance, Yano et al. proposed efficient discrete feature encoding schemes optimized for VQCs, enhancing learning performance while minimizing circuit depth [35].

B. FEATURE EXTRACTION

Quantum machine learning models use different learning paradigms based on how they adapt to data and generate predictions. These methods vary in terms of training mechanisms, quantum-classical interactions, and target outcomes. Three of the most prominent learning approaches are kernel-based learning, variational circuits, and generative modeling.

1) QUANTUM KERNEL METHODS

Quantum kernel methods use quantum circuits to project classical input data into high-dimensional Hilbert spaces through quantum feature maps. Once embedded, the inner products between quantum states are computed using measurement-based techniques such as the Swap Test or Hadamard Test, producing entries for a quantum kernel matrix. This matrix serves as input for classical kernel-based models such as support vector machines (SVMs). The appeal of this method lies in its potential to represent complex, nonlinear boundaries that are difficult to capture with classical kernels. Since the learning happens classically while the quantum device is responsible for kernel evaluation, this

method avoids circuit parameter training and is considered efficient for early NISQ applications [36], [37], [38], [39]. Further quantum deep learning applications may also be implemented using quantum kernel methods [40].

2) VARIATIONAL QUANTUM CIRCUITS

VQCs are hybrid learning models that combine quantum circuits with classical optimization routines. The circuit is constructed with parameterized gates, whose angles are adjusted iteratively by a classical optimizer to minimize a quantum-defined cost function, such as classification error or energy expectation value. Learning occurs through either gradient-based methods (e.g., parameter-shift rule) or gradient-free heuristics like COBYLA or Nelder–Mead. VQCs are especially suited to current NISQ devices because they require relatively shallow circuits, and the optimization remains manageable even under hardware noise [4], [41].

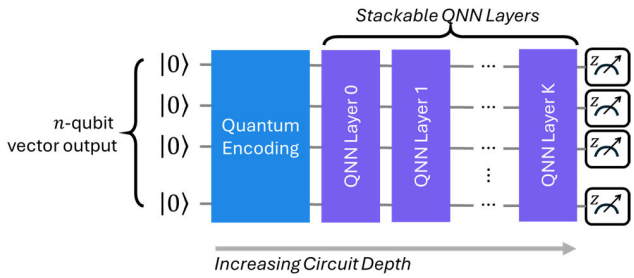


FIGURE 4. Generic QNN circuit.

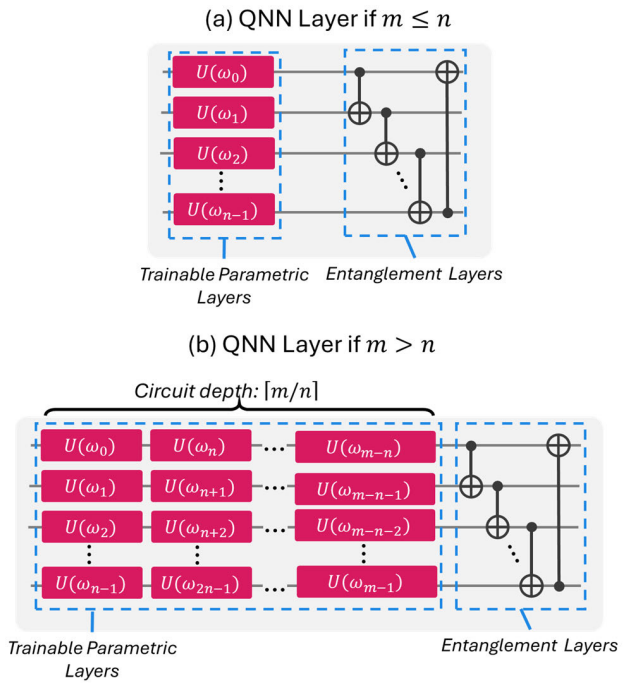


FIGURE 5. QNN layer designs.

QNNs are variational quantum circuits specifically derived from PQCs. QNNs are the direct analogues to ANNs [42], [43], [44] in extracting information from data. However, instead of computing weighted sums followed by nonlinear activations, QNNs evolve quantum states through sequences of unitary operations, where input data is encoded into a quantum state and processed via parameterized quantum gates. A standard QNN circuit typically consists of four main components as seen in Figures 4 and 5: quantum data encoding, trainable layers with parameterized rotation gates, entanglement layers, and measurement gates that enable quantum correlations across qubits. Output is obtained via quantum measurements producing probabilistic results used to compute loss functions.

While ANNs rely on backpropagation and differentiable activations, QNNs optimize parameters using quantum-aware strategies such as the parameter-shift rule or finite-difference estimators. QNNs differ fundamentally in that their representation space is exponentially large (due to Hilbert space), and their behavior is governed by unitary and probabilistic transformations rather than deterministic feedforward logic. Similar to their classical counterparts, QNNs can also have additional hidden layers. This is characterized by increasing the depth of the quantum circuit through modified unitary operations and entanglement circuits. These could also vary depending on hardware where the QNN would be trained on such as using pulse-based hardware [45] or digital quantum computers [40], [46], [47]. Recent works also show techniques in retrofitting classical models onto QNNs using transfer learning [48]. There also exist hybrid methods combining quantum programs and classical machine learnings to improve feature extraction and diversifying resource utilization. Such methods include combining classical SVM with quantum annealing methods [49], VQCs [50], [51], [52] or using dynamic networks [53]. In this review, QNN architectures that rely solely on PQCs for feature extraction are called Pure QNNs. The design of pure QNNs solely depends on the shapes of the feature (input) and target (output) vectors. The feature vector affects the depth of the circuit while target vector shapes determine the number of qubits in a QNN.

C. LEARNING PARADIGMS

Similar to classical machine learning, QML follows the common features of learning paradigms such as supervised, unsupervised, reinforcement, and semi-supervised machine learning.

1) SUPERVISED LEARNING

Supervised learning is the most well-developed application area of QML, where models are trained on labeled data to perform classification or regression tasks. Quantum-enhanced versions of classical algorithms have been explored extensively, especially using quantum kernel methods and VQCs.

Quantum supervised learning models for both classification and regression tasks use either parameterized quantum circuits (e.g., VQCs) or quantum kernel evaluations to learn boundaries in complex feature spaces. Several studies

demonstrate that variational quantum classifiers can outperform classical counterparts on small and noisy datasets by leveraging the high expressiveness of quantum circuits [42], [43], [54], [55], [56], [57]. Recent advances in PQC-based QML models include methods using quantum neurons [43] in various quantum hardware specifically in optics-based systems [42] for hybridizing with established classical methods such as fuzzy logic [57]. QSVMs and quantum-enhanced support vector classifiers also perform well, especially in image and biomedical classification tasks, where the feature space dimensionality benefits from quantum encoding [37], [38].

2) UNSUPERVISED LEARNING

Quantum unsupervised learning aims to uncover structure in unlabeled data using mechanisms fundamentally different from classical methods. While classical models like k-means, PCA, or fuzzy systems rely on iterative numeric computations and distance metrics, quantum models leverage principles like amplitude encoding, quantum state evolution, and inner product estimation. The q-means algorithm [58] replaces classical distance calculations with efficient quantum routines for state comparison, offering potential exponential speedups. Single-preparation strategies alleviate challenges in circuit preparation overhead which allows state reuse to reduce quantum resource demands [59]. Broader surveys outline how quantum Principal Components Analysis (PCA) and clustering operate over compressed, high-dimensional spaces that are intractable classically, while methods like quantum fuzzy regression [60] reinterpret uncertainty modeling through adiabatic Hamiltonians rather than rule-based logic. A more recent addition is the Quantum Self-Organizing Map (QSOM) using Grover's search algorithm, which adapts classical neural mapping to a quantum framework by using amplitude amplification to efficiently identify winning neurons and update neighborhood structures. This approach demonstrates how quantum search primitives can be adapted to model complex data manifolds mirroring the competitive learning process of classical SOMs, but with a potentially faster convergence and richer state encoding [61].

3) REINFORCEMENT LEARNING

Quantum Reinforcement Learning (QRL) merges principles from quantum computing with classical reinforcement learning (RL) to enhance decision-making and control processes in dynamic environments. Unlike classical RL, which relies on iterative updates to value or policy functions using stochastic sampling, QRL exploits quantum parallelism and amplitude amplification to explore state-action spaces more efficiently. Early foundational work in [62] formalized QRL frameworks by integrating quantum operators into policy learning, enabling quantum agents to evolve within a probabilistic state space governed by unitary transformations. More recent advancements, such as curriculum-based QRL for quantum control [63], show that QRL can accelerate learning in complex quantum systems by gradually exposing agents to increasingly difficult control tasks. Additionally, quantum algorithms have been proposed to optimize classical RL training loops [64], using quantum computation to improve sample efficiency and convergence rates. In parallel, Quantum Imitation Learning (QIL) [65] adapts expert behaviors via quantum-enhanced cloning strategies, representing another departure from classical counterparts that rely solely on trajectory matching. In contrast to classical RL, QRL often operates over Hilbert spaces, enables non-classical reward propagation, and presents new opportunities for encoding memory, parallel exploration, and entangled policy updates potentially reducing training overhead and enhancing scalability in environments with large or continuous action spaces.

Expanding from the QML implementation of quantum reinforcement learning, quantum deep reinforcement learning (QDRL) implements the same concept of applying quantum operators for improved learning and feature extraction. Across current implementations, two main approaches are evident: hybrid quantum-classical models and fully quantum circuit-based methods. In hybrid QDRL, PQCs are used as policy networks or value approximators within classical RL loops, with learning driven by classical optimizers via techniques like the parameter-shift rule [63], [66], [67], [68]. These methods often operate in simulation environments and apply to real-world tasks such as robot navigation, quantum control, and grid frequency regulation. In contrast, fully quantum approaches rely on quantum-native representations of the state-action space, quantum feedback loops, and quantum measurements to drive policy learning directly within the Hilbert space [67], [69], [70].

4) SEMI-SUPERVISED LEARNING

Quantum semi-supervised learning (QSSL) leverages quantum circuits to extract structure from partially labeled datasets by embedding both labeled and unlabeled examples into shared quantum feature spaces. Methods such as quantum generative adversarial networks (GANs) [71], kernel-based learners [72], and time series classifiers [73] use quantum-enhanced representations to enable information sharing between data types, often via entanglement or amplitude encoding. These models commonly benefit from parallelism and expressive feature mapping, which allow learning from fewer labeled samples. However, they face similar challenges: circuit depth constraints, sensitivity to noise, and complex data encoding especially for sequential data. Compared to classical techniques, QSSL introduces novel mechanisms for generalization through quantum state manipulation, but its scalability and robustness on real hardware remain active areas of research.

V. QUANTUM DEEP LEARNING

Neural networks are vital in the development of classical deep learning. In quantum deep learning, the counterpart of an artificial neural network is the quantum neural network.

TABLE 1. Summary of Recent Advances in QML Model Architectures.

Learning Paradigm	Feature Extraction Method	Model Architecture	Implementation Type	Type of Quantum Computing	Ref.
Supervised	Quantum Kernel Methods	QSVM	Pure	Gate-based	[39], [40]
		Quantum Kernel + Self-Attention	Hybrid	Gate-based	[76]
	VQA	PQC	Pure	Gate-based	[51], [57], [58]
	Pure QNN	Shallow QNNs	Pure	Gate-based	[43]–[45]
		Deep QNNs	Pure	Gate-based	[46]–[48]
		QCNN	Pure	Gate-based	[77], [78]
		QGAN	Pure	Gate-based	[79], [80]
		Sequential Networks (QGRU, QLSTM)	Pure	Gate-based	[81], [82]
	Hybrid QNN	QNN + ANN	Hybrid	Gate-based	[83]
		QNN + Graph Networks	Hybrid	Gate-based	[84]
		Physics-informed neural networks	Hybrid	Gate-based	[85]
		QCNN	Hybrid	Gate-based	[86]
		QNN + GAN	Hybrid	Gate-based	[87]
		QGAN	Hybrid	Gate-based	[88], [89]
	Simulation-based	Quantum Regression	Pure	Adiabatic	[61]
		Quantum Annealing + SVM	Hybrid	Quantum Annealing	[50]
Unsupervised	VQA	Blind Hamiltonian Parameter Estimation	Pure	Gate-based	[60]
	Oracle-based Methods	Grover's Algorithm + ANN	Hybrid	Gate-based	[62]
Reinforcement	VQA	VQC	Pure	Gate-based	[65]
		VQC	Hybrid	Gate-based	[66]
Semi-supervised	VQA	qSGAN	Pure	Gate-based	[73]
		Quantum Least Squares + SVM	Hybrid	Gate-based	[74]
	Simulation-based	QUBO	Pure	Quantum Annealing	[75]

A summary of recent methods is compiled in Table 1 featuring the different QML learning paradigms as well as QDL architectures.

A. HYBRID CLASSICAL-QUANTUM DEEP LEARNING

Critical bottlenecks arise when designing QNNs for high dimension outputs for classification problems or high-temporal resolution time-series prediction problems. Quantum or simulation hardware limits may constrain the number of qubits that could be used in the QNN. Several workarounds for such technical constraints are to augment the feature extraction capabilities of QNNs with classical deep learning models. Aside from qubit scarcity, limits in quantum circuit depths may affect quality and efficacy of QNNs in feature extraction. Proposed workarounds for such NISQ era challenges are addressed by implementing hybrid quantum neural networks (HQNNs) further illustrated in Figure 6 wherein QNNs are combined with classical deep learning models in a single end-to-end architecture [88].

Classical feature extraction can be carried out by foundational or state of the art deep learning models and QNNs can be placed with or replace fully connected neural network (FCN) layers within the network [89]. QNN placement varies depending on computing and engineering design requirements. QNNs may be placed in between classical FCNs as a dressed QNN [90]. QNNs may be used as an encoder layer [91] or a header layer provided that either input or output vector shapes as illustrated in Figure 7. Having a hybrid quantum architecture allows well known learning techniques such as transfer learning [48]. More advanced implementations of HQNNs include physics-informed neural networks [83].

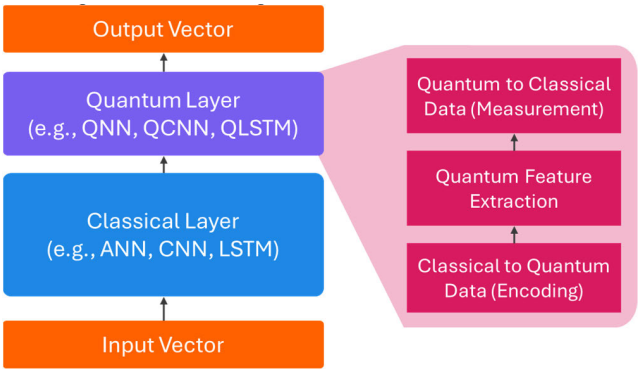


FIGURE 6. Common implementation of a HQNN architecture.

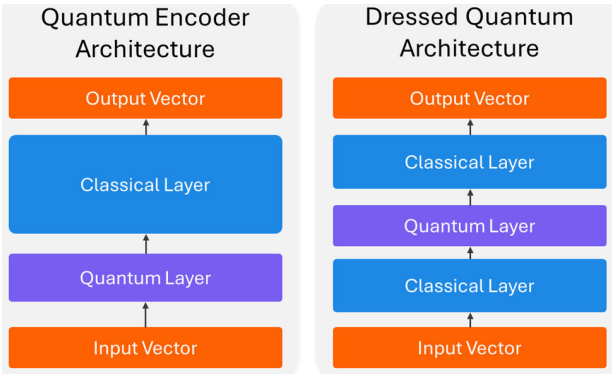


FIGURE 7. Other configurations of HQNNs based on the placement of the QNN layer.

B. QUANTUM CONVOLUTIONAL NEURAL NETWORKS

The quantum convolutional neural networks (QCNN) are also called quanvolutional neural networks in some works. In this review, we consider QCNNs as a pure quantum deep learning operation compared to several previous literature that would refer to QCNNs as a hybrid quantum neural network that combines classical CNNs and QNNs.

One explanation by [75] represents QCNNs akin to Multi-scale Entanglement Renormalization Ansatz (MERA), a tensor network model used in quantum many-body physics to efficiently represent scale-invariant quantum states. The MERA-like structure allows QCNNs to identify global properties from local features and has been successfully used in applications such as quantum phase classification, where learning non-trivial topological or symmetry-protected phases is crucial. Another perspective is explained by [76] further coining the term quanvolution. Quanvolutional operations are quantum analogues to classical convolution filters. Classical convolution kernels are implemented as a PQC and then measured to produce classical outputs. These outputs form a transformed feature map, which is then passed into subsequent neural network layers. Shallow quanvolution circuits can extract nonlinear and high-dimensional features that may not be easily accessible using classical kernels. Importantly, the quantum circuits in a quanvolutional layer are typically not trained during the learning process; instead, their randomness or structure introduces a form of learned quantum feature transformation that acts as a static filter.

C. QUANTUM GENERATIVE ADVERSARIAL NETWORKS

Quantum generative models aim to learn and reproduce the probability distribution of a dataset using quantum circuits. These models generate samples that statistically match target distribution and are analogous to classical generative frameworks like GANs or Boltzmann Machines. Key examples include Quantum Boltzmann Machines (QBMs) [35], [36], and Quantum Circuit Born Machines (QCBMs) [37], and Quantum GANs (QGANs) each leveraging quantum state evolution or amplitude sampling to generate synthetic data.

QGANs extend the classical GAN framework into the quantum domain by training a quantum generator to produce quantum states that mimic a target distribution, while a discriminator attempts to distinguish between real and generated data [77]. QGANs are shown to learn and load random distributions onto quantum states using hybrid quantum-classical loops, highlighting the potential for probabilistic data modeling and state preparation [78]. Enhanced QGAN training stability can be achieved by introducing Rényi divergences in the loss function, replacing classical Jensen-Shannon divergence to better exploit quantum statistical features [86]. QGAN implementations vary depending on implementing a pure quantum generator and quantum discriminator circuit [92]. Hybrid QGANs may implement QNN circuits either in the generator or in the discriminator [85], [86], [87], [93], [94], [95]. These designs share core challenges, including circuit depth limitations, measurement noise, and non-trivial parameter optimization.

D. QUANTUM SEQUENCE NETWORKS

Sequence networks in deep learning paved the path for improved time-series models and natural language-based models such as large language models (LLM). QML has also seen improvements in sequence models. Quantum Recurrent Neural Networks (QRNNs) [79], [96] simulate memory by cycling quantum states through time-dependent gates, allowing sequential inputs to evolve across layers with feedback-like behavior. Extensions like the Quantum Long Short-Term Memory (QLSTM) network [80] emulate gated update mechanisms familiar in classical LSTMs, enabling better retention of long-term dependencies.

Parallel to these developments, quantum analogs of self-attention mechanisms have been introduced to capture inter-token relationships. Quantum Self-Attention [97], [98] Networks compute contextual relevance between quantum-encoded inputs by using operations such as controlled unitaries, entanglement, or inner product estimation (e.g., SWAP tests) to dynamically reweight components based on quantum similarity. These mechanisms allow the model to focus on the most relevant parts of an input sequence, much like classical attention, but within the superposition-rich, high-dimensional quantum feature space.

Quantum transformers adapt the transformer architecture to quantum settings by using PQCs to implement multi-head self-attention, positional encoding, and layer stacking, enabling models to learn contextual dependencies in quantum or classical data [97], [99]. These architectures leverage quantum properties such as entanglement and amplitude encoding to enhance information compression and parallelism, but face challenges like circuit depth, qubit connectivity, and noise sensitivity on current

VI. QML IN APPLIED SYSTEMS AND ENGINEERING

QML is transitioning from theoretical exploration to real-world deployment, with applications emerging across industries and technologies. This section highlights how QML is applied in both discriminative and generative tasks, integrated within technological platforms, and tailored to domain-specific needs in areas like healthcare, energy, and materials science.

A. GENERAL ML ACTIVITIES

Figure 8 presents the distribution of learning paradigms across the works covered in this review. It shows that majority of the applications use supervised learning. Reinforcement learning also has more applied works compared to unsupervised and self-supervised applications combined.

Since supervised learning encompasses regression, classification, and forecasting/estimation tasks, it is also the predominant approach in the machine learning applications observed among the reviewed works, as shown in

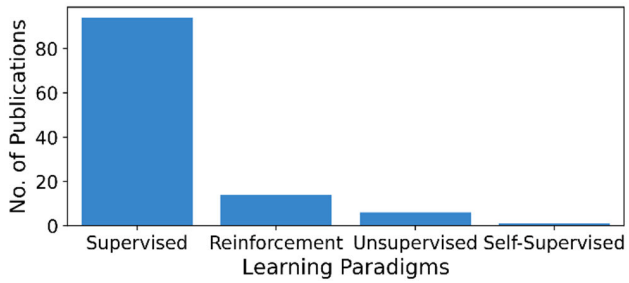


FIGURE 8. Distribution of learning paradigms.

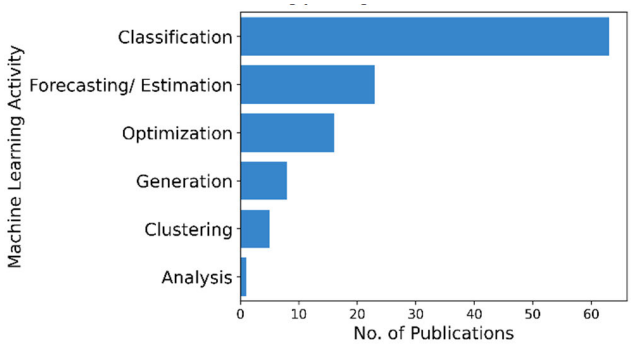


FIGURE 9. Distribution of ML activities.

Figure 9. Optimization tasks have also been addressed through combinations of supervised and reinforcement learning, while generative applications are supported by both supervised and unsupervised learning. Clustering and analysis through unsupervised learning are less common among the reviewed applications. It is also relevant to understand the coverage of the deep learning applications of QML. Figure 10 depicts the distribution of the related works over the scoped period of this review for discriminative and generative artificial intelligence.

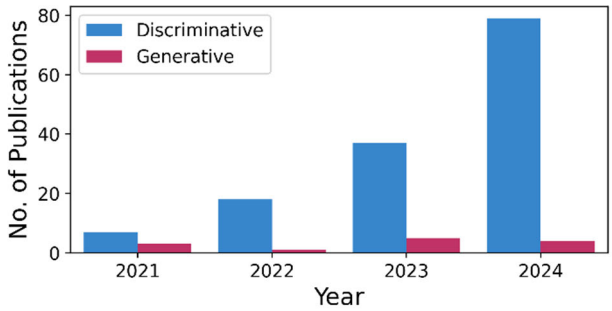


FIGURE 10. Temporal trend of discriminative and generative QML on applied systems.

An exponential trend is seen for QML applications on discriminative AI while there is a fluctuation in the trend in using QML for generative AI (GenAI). This may suggest that the adoption of QML for simpler applications in applied systems is growing in popularity while adoption for GenAI is still in the early adoption stage wherein the NISQ era limits may hinder the development velocity of the technology. The majority of the QML literature in applied systems are discriminative AI. These cover aspects of regression, classification, clustering, and optimization. Table 2 details several works implementing discriminative QML applications given their respective architectures. On the other hand, QML-based GenAI focuses primarily on synthetic data generation and super-resolution applications both in structured and unstructured data. QML used in GenAI applications were done primarily for synthetic data generation. The subsequent discussions will focus on specific discriminative and generative QML applications in two aspects: (a) technical and engineering methods and (b) industrial and scientific applications.

B. TECHNICAL AND ENGINEERING METHODS

One perspective in assessing QML applications is looking at them as a method for achieving engineering applications and industrial applications.

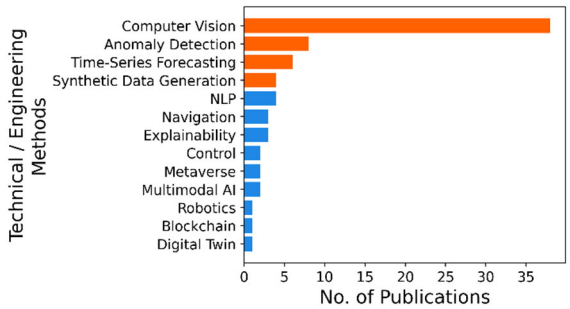


FIGURE 11. Distribution of technical/engineering methods.

Figure 11 depicts the role of QML in various technical and engineering methods that could be essential parts in an applied system.

1) COMPUTER VISION

Quantum machine learning is increasingly applied to computer vision tasks, utilizing hybrid and variational quantum circuits to address challenges in classification, recognition, segmentation, and generative modeling. A recurring architecture across many works is the VQC-QCNN, used for both image classification and handwritten text recognition, often employing entangling gates for feature extraction [100], [101], [102]. Several studies use hybrid quantum-classical generative adversarial networks for image super-resolution and generation, blending classical discriminators with quantum generators to improve detail reconstruction in discrete image distributions [103], [104]. Quantum feature extraction is another central theme, implemented using single-qubit encodings, quantum Fourier transforms, or dense locality-aware encodings for object counting, segmentation, and face recognition [105], [106], [107], [108], [109]. In unsupervised and weakly supervised tasks, methods such as inverse QFT-based segmentation and automatic QML optimize architectures via evolutionary algorithms [110], [111]. While unsupervised techniques can also be applied in enhancing similarity learning [112] which are helpful for facial recognition applications.

2) ANOMALY DETECTION

The second most common application of QML as a technical or engineering method for applied systems is anomaly detection. Quantum feature selection or encoding is used alongside classical classifiers to enhance detection sensitivity [113], [114]. VQCs are widely employed to model complex non-linear feature spaces, providing computational advantages for highly imbalanced datasets like financial fraud cases [115]. Quantum autoencoders also compress normal data patterns into low-dimensional quantum states, making anomalies easier to detect through reconstruction errors [116]. Quantum annealing simulations have also been utilized for predictive tasks like pause location prediction, optimizing combinatorial aspects of the anomaly detection workflow [117].

3) TIME-SERIES FORECASTING

QML is also gaining traction in time-series forecasting applications. A dominant trend across recent works is the use of hybrid quantum-classical architectures, where quantum-enhanced layers such as Quantum Long Short-term Memory (QLSTM) or quantum Gate Recurrent Unit (GRU) variants are integrated into deep learning models to capture non-linear temporal patterns [118], [119], [120], [121]. VQCs or quantum gates are commonly employed to encode dynamic dependencies within sequences [122]. Optimization techniques such as particle swarm algorithms are further combined with quantum models to enhance training robustness in noisy environments [123]. Across these studies, quantum models are shown to improve generalization, short-term prediction accuracy, and model compactness when compared to purely classical counterparts, though practical deployment still faces challenges regarding hardware scalability and noise resilience.

4) SYNTHETIC DATA GENERATION

Generative QML applications focus mainly on synthetic data generation. Common models used for generation of data are hybrid quantum GAN architectures. By embedding quantum kernels or parameterized circuits into GAN frameworks, models such as the hybrid quantum–classical GAN [103], Hybrid Quantum Deep Convolutional GANs [124], and quantum conditional GAN [95] aim to leverage quantum entanglement and superposition to enhance diversity and realism in synthetic outputs. Another shared trend is the focus on domain-specific data especially biomedical signals (like ECG waveforms) and remote sensing images, where traditional generative models often struggle with sparse or irregular patterns [95], [124]. Techniques like patch-based learning [95] and quantum feature enhancement [125] further demonstrate a movement towards localized generation strategies, improving resolution and structural fidelity.

5) OTHER APPLICATIONS

QML applications can also be extended in metaverse [126], [127] applications and digital twins [128]. Automation improvements have also been on the rise as QML can be used in system control [67], [70] and navigation learning [66] in robotics specifically in autonomous aerial vehicles [129], [130]. Works in multimodal QML [131], [132] already have been developed which can be melded with NLP [133] applications such as for sentiment analysis [134], [135], [136]. Initial efforts in the field explainable data science and AI have been developed [132], [137].

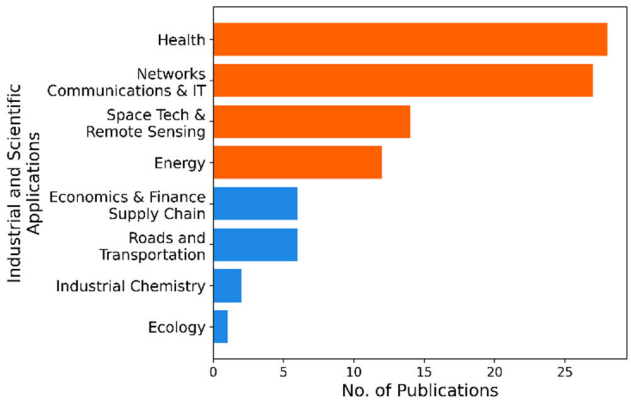


FIGURE 12. Distribution of industrial and scientific QML applications.

C. INDUSTRIAL AND SCIENTIFIC APPLICATIONS

The realization of QML in industrial and scientific applications is essential for its widespread adoption. Figure 12 shows the roles of QML in several industry specific domains across the scope of this review.

1) HEALTHCARE

QML applications in the healthcare domain are the most prevalent among all industrial and scientific applications, particularly where pattern recognition and limited data availability hinder traditional machine learning. In cardiology, quantum-enhanced classifiers have shown efficacy in interpreting complex biological signals like ECGs and detecting ischemic events [138], [139]. Oncology and neurology benefit from quantum models' ability to extract latent spatial and temporal features from modalities such as electroencephalograms (EEGs) and Magnetic Resonance Imaging (MRI), aiding early tumor or neurodegenerative disorder detection such as Alzheimer's disease [140], [141], [142], [143], [144], [145]. In medical imaging, hybrid quantum–classical convolutional models have improved segmentation and classification accuracy in X-rays, thermal imaging, and retinal scans by leveraging quantum circuits' richer feature encoding [146], [147], [148], [149], [150]. RGB-based imaging is also used in dermatological applications [151]. Pharmaceutical research has employed quantum generative models for protein design and molecule synthesis, where quantum systems natively simulate molecular interactions [94], [152], [153]. Microbiology and Pathology diagnostics are also beginning to adopt quantum-enhanced classification techniques for bacterial identification and pathogenic pattern

discovery [131], [154]. Psychiatric and mental health related applications of QML then primarily focus on recognition of preemptive and analysis of patients [155], [156]. Moreover, QML can be implemented in general health management and prognostics, such as machine health monitoring and sustainable healthcare edge optimization, and beyond conventional diagnostics [157], [158], [159], [160], [161].

Hybrid quantum-classical frameworks are mostly used in health applications, combining classical neural networks (e.g., CNNs, LSTMs) with quantum submodules such as QCNNs, QSVMs, and VQCs [138], [141], [147], [160], [162]. These models leverage quantum properties such as entanglement and high-dimensional Hilbert spaces for improved feature encoding, compression, and parallelism. QCNNs dominate medical imaging and multimodal fusion tasks [124], [146], while QGANs and quantum autoencoders address synthetic data generation challenges in pharmaceutical and cardiology datasets [94], [124]. In neurocognitive diagnostics, QSVMs and QNNs offer scalable architectures for early classification with limited training data [141], [159].

2) NETWORKS, COMMUNICATIONS, AND IT

Quantum machine learning has emerged as a versatile tool in a variety of networking subdomains, particularly in enhancing adaptability, efficiency, and security. In edge, IoT, and cloud systems, QML supports decentralized learning and resource-aware intelligence, enabling energy-efficient processing, federated model updates, and scalable deployment on edge devices [157], [163], [164], [165]. QML applications in traffic management and optimization quantum-enhanced reinforcement learning techniques facilitate more dynamic allocation of network resources, enabling improved throughput, latency management, and load balancing in complex environments [68], [163], [164], [166], [167], [168]. Network security also leverages QML for advanced anomaly detection, privacy-preserving data handling, and secure data classification [169], [170], [171], [172], [173], [174], [175], [176]. Broader applications in general network intelligence focus on network diagnosis and adaptive learning frameworks for intelligent communications infrastructure planning [126], [177], [178], [179], and blockchain systems [161], [180]. Security systems, specifically surveillance systems, can also use QML for anomaly detection [181], [182]. Finally, software development management can also be improved by QML for software validation [183].

QNNs are widely used in classification and resource control tasks, offering a balance between expressiveness and quantum efficiency [171], [174], [178], [184]. QRL is especially prevalent in dynamic environments, where sequential decision-making and policy optimization are critical [68], [126], [165], [167]. In contrast, QSVMs are employed in fault diagnosis and high-dimensional classification with interpretable model behavior [172], while quantum clustering and privacy-preserving methods are utilized for secure and unsupervised learning tasks [169]. Grover's algorithm continues to find utility in optimizing search-based anomaly detection systems [165], [173], and QAOA-based optimization methods are applied in sustainable edge infrastructure and intelligent service deployment [157]. Finally, hybrid quantum deep learning (Hybrid QDL) remains a flexible design approach that embeds quantum layers into classical deep networks for enhanced learning fidelity specifically hybridizing using QNNs [161], [177], [179] and QCNNs [175], [185].

3) SPACE TECHNOLOGY AND REMOTE SENSING

Another key domain in which QML has garnered attention is space technology. In remote sensing, QML methods are widely employed to enhance satellite image classification and regression, particularly under data-limited and high-dimensional conditions. Hybrid quantum neural networks and QUBO-based kernel approximations have successfully handled multispectral image data and real-world remote sensing tasks [186], [187], [188], [189], [190]. In earth observation, QML techniques are applied to hyperspectral image classification, land cover mapping, and atmospheric condition detection. Hybrid quantum-classical convolutional and graph neural networks are used to analyze satellite image structures and detect dynamic changes, while quantum kernel methods provide effective classification under limited data [125], [191], [192], [193], [194]. In the exploration domain, QML facilitates control and trajectory optimization of space vehicles and autonomous aerial vehicles (AAVs) through quantum-enhanced reinforcement learning strategies [129], [130], [195]. Finally, in astrophysical applications, QML aids in the classification of celestial objects using supervised quantum SVM models, supporting early detection and hazard identification [196].

Hybrid QCNNs dominate in multispectral and hyperspectral image classification, especially in remote sensing and earth observation, where spatial detail and low sample availability are critical factors [125], [186], [187], [192]. In regression and feature selection tasks, QUBO-based methods [188], [194] implemented on quantum annealers provide efficient solutions for subset selection and kernel approximation [197] reducing computational cost while maintaining model fidelity. QNNs [189], [190] and graph neural networks (GNNs) [194] are used in more complex spatial learning scenarios, such as change detection or trajectory planning, where structured inputs and relational dependencies are critical. For dynamic optimization and control problems, QDRL and layer-wise hybrid RL architectures are applied, showcasing adaptability in pathfinding and space landing scenarios [129], [195].

4) ENERGY

QML has been applied in the energy sector particularly in areas requiring predictive modeling, control optimization, and decision automation. In the domain of renewable energy forecasting, quantum-enhanced models have been applied to short-term and day-ahead prediction tasks for solar irradiance

and wind speed, employing hybrid quantum-classical architectures, QLSTM, and gated recurrent quantum units [118], [119], [120], [121], [123], [198]. The energy control and grid operations category encompass works on clustering-based energy profiling, and quantum reinforcement learning for frequency regulation and dynamic stability in distributed systems [67], [70], [128]. Meanwhile, energy trading and economics explores quantum-driven strategies for market behavior modeling and decentralized decision-making, as demonstrated in quantum RL frameworks designed for blockchain-based electric mobility trading systems [180].

Hybrid quantum-classical architectures are predominant forecasting and control-intensive applications in the energy domain. Models such as QLSTM and QGRUs are frequently used in hybrid configurations for time-series prediction of solar irradiance and wind speed, benefiting from both classical sequential learning and quantum-enhanced state representation [119], [120], [123]. Similarly, QNNs appear across a variety of forecasting [118], [121] and hybrid control models [122], often integrated with classical reinforcement strategies to manage distributed systems or optimize power flow. Pure and hybrid QDRL models are applied in tasks like rotor-side converter optimization and frequency control [67], [70], [180], while Quantum k-means clustering and Grover's algorithm are explored in unsupervised operation modeling and decentralized energy trading [180], [199].

5) OTHER INDUSTRY DOMAINS

Other industry domains such as in transportation and mobility have also applied QML techniques for forecasting, anomaly detection, and classification for mobility and transportation planning [117], [201], [202], [203], traffic management [204], and anti-theft purposes [114]. Economic and financial applications use QML in areas such as fraud assessment [113], [115], [116], stock price prediction [132], [205], and supply chain management [206]. A few applications can also be seen in industrial chemistry [207], [208] and in ecological applications [209].

VII. INFRASTRUCTURE AND OPERABILITY

Another important factor in applied systems is the infrastructure and deployment strategies of developed programs. In this section, implementation and operability strategies across works are reviewed.

A. IMPLEMENTATION

Strategies for implementing systems vary from use cases, ML activity, or ensemble of models in a developed system. Different computational requirements for both hardware and software resources are also considered in the works.

1) QML WORKFLOWS

Development and deployment pipelines or workflows are necessary for systems engineering and realizing applied QML systems in their industrial domains. QML workflows are similar to classical machine learning development and ML operation (MLOps) pipelines in the aspect of data loading, training, optimization, and possible deployment strategies in the general sense.

2) COMPUTATIONAL REQUIREMENTS

The number of qubits required for each technical/engineering method and industrial/scientific application depends on the QML method and utility requirements.

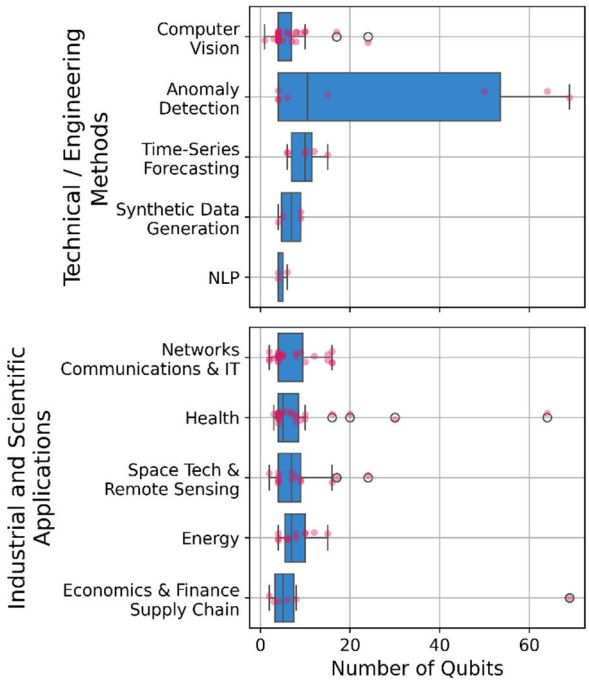


FIGURE 13. Qubit counts per technical/engineering method and industrial/scientific applications.

Figure 13 shows the different qubit utilization for the works in this review. There is a seen cluster of qubit utility for technical and engineering methods with a round 4 qubits. This is primarily due to the default configuration of quanvolution operations specifically with Computer Vision applications. Such methods mainly use QCNN architectures similar Synthetic Data Generation applications using Hybrid QGANs.

Anomaly detection has a wider variance in qubit utilization, considering implementations using QUBO models, which can cater to more qubits with the D'Wave quantum annealer than other digital quantum computer alternatives. Time-series forecasting qubit utilization is distributed from 6 to 15 qubits, reflecting either the number of qubits for encoding or the output vector size. For applied industrial and scientific applications, the common qubit utility for a hybrid QCNN or QNN applications with CNN, ANN, and GANs starts from 4 qubits, as seen in Figure 14.

Figure 13 further shows that Networks, Communications, IT, Health, and Space Technologies, and Remote Sensing applications in which most of their applications have QCNN and QNN implementations. Energy and Roads/ Transportation applications have sparse distribution of qubit utility as

TABLE 2. Summary of recent QML industrial and scientific applications.

Industry or Scientific Domain	Sub-domain	Type	Quantum Method	Classical Method	No. of Qubits	Hardware	Platform	Ref.
Healthcare	Cardiology	Pure	QNN	-	30	CPU	Qiskit	[157]
		Pure	QGAN	-	9	GPU	Pennylane	[98]
		Hybrid	QNN	CNN	20	CPU	Qiskit	[141]
		Hybrid	QCNN	CNN	4	IBM Q, GPU	Qiskit, Pennylane	[142], [149]
		Hybrid	QCNN	GAN	4	GPU	Pennylane	[127]
	Neurology	Pure	QNN	-	-	GPU	MATLAB	[143]
		Pure	QSVM, Quantum Wavelet Transform	-	7	CPU	Qiskit	[203]
		Pure	QCNN	-	8	-	-	[147]
		Hybrid	QNN	ANN	4	GPU	Pennylane	[144]
		Hybrid	QSM	CNN	6	GPU	Qiskit	[148]
	Oncology	Pure	QNN	-	-	GPU	MATLAB	[143]
		Pure	QSVM	-	3	CPU	Qiskit	[145]
		Hybrid	QCNN	CNN	4	GPU	Qiskit	[146]
	Pharmaceutics	Pure	QSVM	-	8	IBM Q	Qiskit	[156]
		Hybrid	QGAN	GAN	10	IBM Q	Qiskit	[97]
		Hybrid	QNN	ANN	64	CPU	Mathematica	[155]
	Psychiatrics	Hybrid	QNN	LSTM	8	CPU	Qiskit	[159]
		Hybrid	QCNN	CNN	4	GPU	Pennylane	[158]
	Rheumatology	Hybrid	QSVM	k-means	7	IBM Q	Qiskit	[152]
	Pathology	Hybrid	QCNN, QNN	GAN	4	GPU	-	[134]
	Dermatology	Hybrid	QCNN, QSVM	CNN	4	CPU	Pennylane, Qiskit	[154]
	Ophthalmology	Pure	QSVM	-	10	CPU	Qiskit	[151]
	Health Management	Hybrid	QAOA	CNN	16	GPU	-	[160]
		Hybrid	QNN	ANN	5	GPU	Tensorflow Quantum	[162]
	Generic	Hybrid	QCNN	CNN	4	CPU	Pennylane, Qiskit	[150], [163], [165]
Networks, Communications, and IT	Vulnerability Assessment	Pure	Grover's Algorithm	-	15	-	-	[176]
		Pure	QNN	-	2, 8	CPU, IonQ	- , Cirq	[174], [177]
		Pure	QSVM	-	2	IBM Q	Qiskit	[175]
		Hybrid	QCNN	CNN	4	-	-	[178]
	Traffic management and optimization	Pure	QNN	-	9	IBM Q	Qiskit	[187]
		Hybrid	QNN	Federated Learning	10, 8	GPU, CPU	Torchquantum, Qiskit	[69], [166]
		Hybrid	QNN	Deep RL	4,	IBM Q, CPU	Qiskit	[169], [170]
	Edge, IoT, and Cloud Systems	Hybrid	QNN	Deep RL	15	CPU	Qiskit	[167]
		Hybrid	QNN	ANN	2	IBM Q	Qiskit	[164]
		Hybrid	Grover's Algorithm	Deep RL	-	GPU	-	[168]
		Hybrid	QAOA	CNN	16	CPU	-	[160]
	Network Infrastructure and Operations Planning	Pure	QNN	-	5, 3	CPU	- ,Pennylane	[171], [181]
		Hybrid	QNN	ANN	16	GPU	TensorCircuit	[182]
		Hybrid	QNN	CNN	4	GPU	Pennylane	[180]
	Privacy	Pure	QNN	-	5	IBM Q	Qiskit	[172]
		Hybrid	QNN	CNN	4	GPU	Qiskit	[173]
		Hybrid	QCNN	LSTM	12	GPU	Cirq	[188]
	Blockchain	Hybrid	Grover's Algorithm	RL	4	Rigetti	PyQuill	[183]
		Hybrid	QNN	ANN	2	IBM Q	Qiskit	[164]
	Surveillance Systems	Hybrid	QNN	CNN	4	GPU	Qiskit	[184]
		Hybrid	QCNN	CNN	4	-	Pennylane	[185]
	Software Validation	Hybrid	QNN	ANN	4	GPU	Torchquantum	[186]
Space Technology and Remote Sensing	Remote Sensing	Hybrid	QCNN	CNN	4	IBM Q	Qiskit	[190]
		Hybrid	QNN	CNN	4	GPU	Pennylane	[192]
		Hybrid	QNN	ANN	17	GPU	Tensorflow Quantum	[193]
		Hybrid	QUBO	Support Vector Regression, Gaussian Process Regression	16	Quantum Annealer	D'Wave	[191]
	Earth Observation / Aerial Imaging	Pure	QSVM	-	24	CPU	Pennylane	[194]
		Hybrid	QCNN	ANN	7	GPU	Tensorflow Quantum	[195]
		Hybrid	QCNN	GAN	9	GPU	-	[128]
		Hybrid	QNN	GNN	4	GPU	Pennylane	[197]

TABLE 2. ***(Continued.)*** **Summary of recent QML industrial and scientific applications.**

		Hybrid	QUBO	Decision Tree	9	Quantum Annealer	D'Wave	[196]
	Exploration-related	Hybrid	QDRL	DRL	2, 4	- , IBM Q	- , Qiskit	[132], [198]
	Astrophysical	Pure	QSVM	-	7	IBM Q	Qiskit	[199]
Energy	Renewable Energy	Hybrid	QNN	LSTM	10	GPU	Pennylane	[121], [122]
		Hybrid	QLSTM	ANN	12	GPU	Pennylane	[123]
		Hybrid	QGRU	ANN	60	GPU	Pennylane	[126]
		Hybrid	QNN	CNN	6	GPU	CUDA-Q	[124]
	System Control	Hybrid	QNN	ANN	6	-	-	[131]
		Hybrid	QNN	DRL	8	GPU	Pennylane	[68]
		Hybrid	QDRL	Deep Belief Networks	4	CPU	MATLAB	[71]
	Operations and Simulation	Pure	Quantum k-means	-	8	IBM Q	Qiskit	[202]
		Hybrid	QNN	ANN	15	-	-	[125]
	Energy Trading	Hybrid	Grover's Algorithm	RL	4	Rigetti	PyQuil	[183]
Roads and Transportation	Transportation and Mobility	Pure	Quantum Self-Attention Models	-	5	CPU	Qiskit	[204]
		Hybrid	Quantum Bayesian Networks	LSTM	10	CPU	Qiskit	[205]
		Hybrid	QUBO	CNN	50	Quantum Annealer	D'Wave	[120]
		Hybrid	QUBO	DRL	6	Quantum Annealer	D'Wave	[206]
	Traffic Management	Pure	QCGNN	-	10	GPU	Tensorflow Quantum	[207]
	Anti-theft	Hybrid	QUBO	Restricted Boltzmann Machine	64	Quantum Annealer	D'Wave	[117]
Economics, Finance, and Supply Chain	Fraud Assessment	Pure	Quantum Autoencoder	-	4	IBM Q	Qiskit	[119]
		Pure	QNN	-	6	IBM Q	Qiskit	[118]
		Hybrid	QSVM	XGBoost	69	CPU	Qiskit	[116]
	Stocks	Hybrid	QNN	ANN	8	-	-	[208]
		Hybrid	QNN	Multimodal	3	GPU	Qiskit, Pennylane	[135]
	Supply Chain	Hybrid	QNN	ANN	2	GPU	Pennylane	[209]
Industrial Chemistry	Chemistry	Pure	QSVM, QCNN	-	8	CPU	Qiskit	[210]
	Manufacturing Process	Pure	QSVM	-	8	CPU	Qiskit	[211]
Ecology	Marine Biology	Hybrid	QNN	CNN	10	GPU	Pennylane	[212]

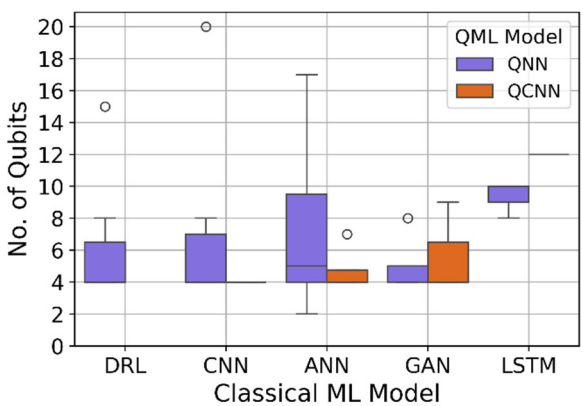


FIGURE 14. **Qubit utilization for hybrid models specifically for QNN and QCNN models.**

this is related to optimization applications. Several apppli-cations would require different number of qubits based on optimization variables in QUBO or QAOA models implemented in reinforcement learning implementations.

Understanding qubit utilization provides scientists and engineers with usable qubit estimates based on prior literature or industrial requirements. However, it may also be helpful for quantum computer hardware and software platform providers to have an idea of the qubit requirements of possible practical applications or constrained positions of applied systems research work.

3) HARDWARE

The infrastructure needed for running applied systems is vital for future operations. Understanding the current hardware utilization would be helpful to determine the needs and constraints of recent works. Figure 15 shows the overview of the different hardware in the works covered in this review.

However, it is noted that data points for Figs. 14-16 only include papers that declared the hardware utilized. Actual quantum computing devices such as IBM Q, D'Wave, Rigetti, Strawberry Fields, and IonQ may be used through cloud access. However, most works for technical/engineering methods and applied systems use classical simulators through

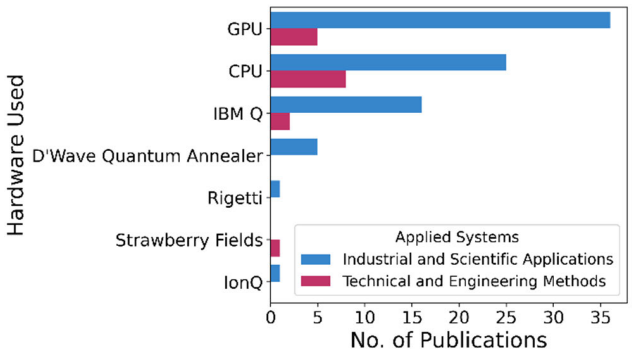


FIGURE 15. Hardware used for applied systems.

GPUs and CPUs. This may imply that most works use high-performance computing devices to simulate and design quantum algorithms in applied systems. The preference for using classical hardware may lie in the research development turnaround time, accessibility of quantum hardware alternatives, and suitability of the designed systems with existing quantum hardware.

Hardware used for different industrial and scientific applications is shown in Figure 16. The top domain applications predominantly use classical hardware, indicating initial reliance on HPCs for prototyping research or constraints in accessing quantum hardware. Another factor is the use of quantum software platforms such as Pennylane, Tensorflow Quantum, and CUDA-Q, providing convenient means for researchers to develop, test, or prototype QML applications using CPUs and GPUs to simulate quantum circuits and train hybrid quantum models.

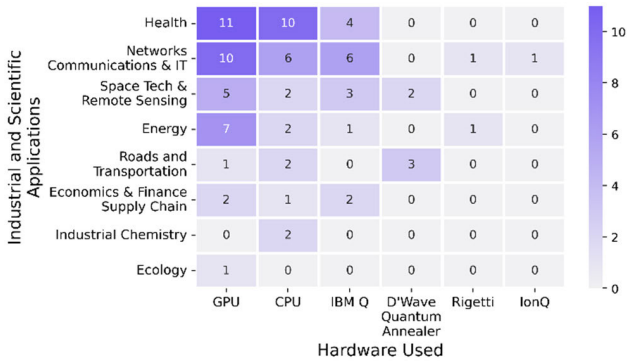


FIGURE 16. Hardware usage for each industrial/scientific application.

B. OPERABILITY

Current alternatives in the literature emphasize hybrid quantum–classical workloads and distributed resource utilization to overcome the limitations of the NISQ era. These constraints directly impact the design and deployment of QML/QDL workflows and are central to the emerging field of QMLOps. Two key operational bottlenecks are often highlighted. First, variational computation costs: quantum hardware is primarily accessible via cloud platforms, where costs accumulate per job submission. As parameterized quantum circuits (PQCs) must be executed at each training step, the computational cost scales with the number of training epochs. Second, communication latency: model training can be hindered by delays in data exchange between quantum and classical components. Synchronization, data conversion, and remote execution introduce nontrivial overhead. The following methods from prior works are proposed to remedy these bottlenecks.

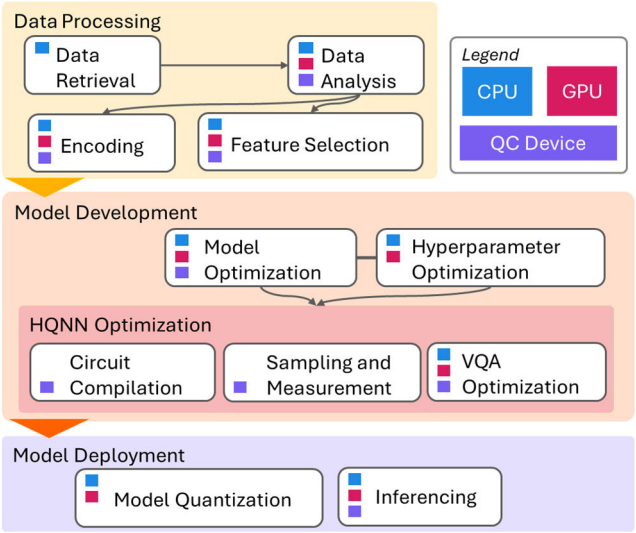


FIGURE 17. Generic QML workflow with dedicated hardware use.

1) HYBRID WORKLOADS

Utilizing both computational powers of HPCs and quantum processing units would solve challenges in quality and qubit scarcity. Specifying different hardware for specific MLOps activities would improve general operations. A QML workflow introduced in the covered works that could maximize use of existing resources is prototyping in classical hardware first then implementing on available quantum hardware. This may also be improved using transfer learning [48], [189] and knowledge distillation [210], [211], [212].

A QML workflow is presented in Figure 17, highlighting the applicable hardware for both classical and quantum machine learning activities. Data processing, basic data exploration, data wrangling, data loading activities, or optimization in VQA routines can run on CPUs. GPUs can run classical deep learning operations. Deep quantum circuit simulations in HQNNs can be run on GPUs in the absence of quantum hardware. Quantum computing devices may then be used for quantum workloads such as quantum encoding, feature extraction, and simulation of quantum circuits [121].

2) DISTRIBUTED UTILITY

Although the use of diverse hardware may increase total efficiency but may not improve total throughput. Fig. 18 (a) shows the local HPC implementation wherein CPU, GPU and quantum devices are aggregated in a computing cluster. However, significantly high capital expenditure costs for

each institution are expected. Distributed computing efficiently allocate dedicated jobs or workloads for available resources [41], [213]. Economies of scale in cloud-hosted HPCs, like Fig. 18 (b), can further benefit from such implementation. QML models in large systems can also be broken down into dedicated smaller models. The configuration in Fig. 18(c) shows the coordination of different quantum devices where quantum optimization is on annealing devices, gate-based operations on QPUs, and variational optimization on classical hardware. Such implementations are evident in QDRL applications. This implementation follows federated machine learning techniques [51], [51], [128], [163], [213] that could allow distillation and synchronization of model weights over diversified and dedicated hardware similar to configuration Figure 18 (d).

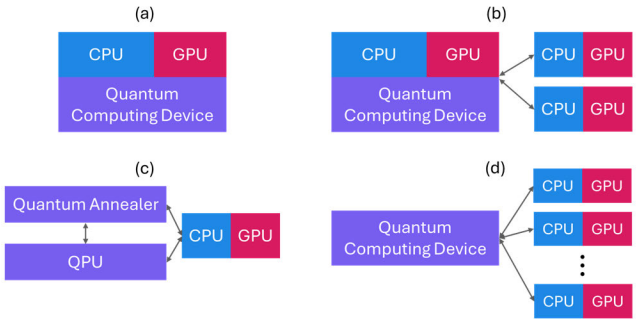


FIGURE 18. Hybrid classical quantum system deployment strategies.

C. LIMITS OF COMPUTATION: CLASSICAL ML VERSUS QML

The primary difference between classical and quantum machine learning lies in the reliance on quantum mechanics and quantum information theory for data representation, learning, and optimization—leveraging key quantum resources such as superposition and entanglement. The general and theoretical advantages of quantum algorithms over classical algorithms, as identified in prior works, are summarized as follows:

1. **Data Representation.** Superposition allows data originally stored in classical memory to be represented using fewer qubits.
2. **Reduction in Time Complexity**. Quantum algorithms can achieve improved execution time through entanglement and superposition.

However, it is important to note that, as of the writing of this paper, there is no consensus or universal acceptance on the clear advantage of quantum machine learning over classical machine learning in the early development of the field. Most of the prior works reviewed claim shorter training times, improved prediction accuracy, and increased robustness in learning. Although prior work reports such improvements, these are not sufficient to substantiate a claim of "quantum advantage" in machine learning, as they may only be applicable for specific datasets, curated cases, or niche applications, and not for generalized feature learning [214], [215].

Specific limitations can still be seen in the pure and hybrid QML and QDL methods and practical applications in works featured in this review.

1. *Scalability*. Prior work had little focus on increasing the number of information representations of features or the use of big data. The lack of scalability tests or further experiments may be a critical limitation on the wide-scale implementation of applications.
2. *Cross-compatibility with quantum hardware.* It would be difficult to ascertain the improvements and robustness of QML as an application in real quantum hardware since most of the work primarily simulates QML, QDL, and HQNNs in classical hardware like CPUs and GPUs. Few prior works declared or reported their use of quantum hardware for large QML cases.
3. *Noise resilience.* Although most applied works featured in this review implement gate-based quantum computing, only a few considered noise modelling, error correction, or noise mitigation during optimization. Noise propagated from quantum hardware may critically impact model learning.

VIII. ETHICAL AND RESPONSIBLE QML

Modern AI applications are subjected to responsible and ethical use. As a sub-topic of AI and considering its applications in human society, it is best to understand the scope of ethics and responsibilities in integrating QML into applied systems. Several aspects such as responsible use, regulation, and its socio-economic implications are necessary to ensure the safety and ethics of this emerging technology. In this discussion, concepts from responsible AI literature and responsible quantum technology literature are considered to conceptualize responsible QML integration in applied systems and systems engineering.

The immediate consideration for responsible QML is whether it affects the safety of applied systems involved or used by humans. Physical safety can be rooted with the quality, timing, accuracy, and precision of systems. Considering the limits of QML in the NISQ era, the quality at which it can operate in safety-critical situations such as in disaster risk reduction and healthcare remains ongoing work. Although there are existing standards on AI for user safety [216] and for other quantum technologies [217], there are no specific QML standards as of the writing of this paper that would set safety standards of users. Trustworthiness of AI in handling data used for training or for inference is another concern at which QML may also be faced. An implied feature of QML models is that they cannot be easily tapped and interpreted by any third party unless they have a similar quantum computing device that can decode quantum data. However explicit methods exist to characterize privacy in QML algorithms [218].

The socio-economic considerations in QML may also be considered beyond as these potential applications affect economies and way of life. Since quantum technology is relatively new and has a niche market, the risk of technology monopoly in terms of intellectual property should be avoided

or controlled [219]. This also stems from AI applications that may or someday have advantages due to quantum computing. Risk of disinformation of applications and use of QML can occur if the public does not fully comprehend the technology. The lack of understanding may lead to false expectations or distrust of the technology which in turn hinder innovation and adoption [220]. A general trend of democratizing technology would help decrease ambiguity and reluctance in using the technology [221], [222].

The long-term regulation of QML systems would be considered a similar trend to or earlier than classical AI. Aside from technical standards, proactive governance both in the political and legal level should be considered [223], [224]. Ethical protocols should also be placed aside from legal considerations. This may necessitate professional organizations at various levels (e.g., international, regional, local) to develop codes of conduct in developing, integrating, or using QML in AI pipelines and systems in general ensuring safety and innovation [225], [226], [227], [228].

IX. LIMITATIONS

Although comprehensive, this review is temporally limited to publications from 2021 to 2024. Additional innovations are continuously being developed and published at the start of 2025 and may significantly change the rankings of the industrial and technical applications sectors in this review. Initial concepts and points for QMLOps and Responsible QML are compiled and reported in this review. While proposing technical implementations and ethical guidelines is beyond the scope of this review, future researchers are encouraged to build upon these findings. This review is further limited in providing a comprehensive account of the hardware and qubit requirements across all studies, as many prior works do not disclose their implementation setups or explicitly state the number of qubits used during experimentation.

X. CONCLUSION AND FUTURE WORK

This paper presents a review of the current landscape of QML within the context of applied systems and engineering. By surveying 172 studies across multiple domains, we analyzed the theoretical underpinnings, algorithmic frameworks, and hardware requirements that shape QML development. This review focused on examining the core components of QML, such as data representation, feature extraction, and learning methods, as well as a detailed assessment of quantum deep learning architectures, training strategies, and infrastructure requirements.

Analysis of the current works shows that QML offers promising enhancements in efficiency, scalability, and expressiveness for applied systems dealing with complex, high-dimensional data. HQNNs and VQCs have demonstrated competitive performance in classification, forecasting, and optimization tasks, often under constrained data and resource environments. Notably, QML has been applied to medical imaging, anomaly detection, smart grid control, network security, and satellite data processing, demonstrating a breadth of utility across sectors. Despite ongoing hardware limitations in the NISQ era, the growing ecosystem of quantum software frameworks and simulators facilitates the exploration of QML in mission-critical and engineering-centric domains. The field now stands at the intersection of technical advancement and practical integration, with responsible development, accessibility, and system-level design playing crucial roles in its continued adoption.

This work can be further improved by expanding search areas, such as including non-QML industrial applications, such as quantum optimization, quantum communications, and quantum sensing, for a broader perspective of industrially applicable quantum technologies. Further guides, policies, and frameworks may be further developed based on the qubit requirements, QMLOps, and ethical QML.

REFERENCES

[1] A. J. Daley, I. Bloch, C. Kokail, S. Flannigan, N. Pearson, M. Troyer, and P. Zoller, "Practical quantum advantage in quantum simulation," *Nature*, vol. 607, no. 7920, pp. 667–676, Jul. 2022.

[2] N. Pirnay, V. Ulitzsch, F. Wilde, J. Eisert, and J.-P. Seifert, "An in-principle super-polynomial quantum advantage for approximating combinatorial optimization problems via computational learning theory," *Sci. Adv.*, vol. 10, no. 11, p. 5170, Mar. 2024.

[3] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum machine learning," *Nature*, vol. 549, no. 7671, pp. 195–202, Sep. 2017.

[4] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, Ł. Cincio, and P. J. Coles, "Variational quantum algorithms," *Nat. Rev. Phys.*, vol. 3, no. 9, pp. 625–644, Sep. 2021.

[5] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta, "Supervised learning with quantum-enhanced feature spaces," *Nature*, vol. 567, no. 7747, pp. 209–212, Mar. 2019.

[6] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, "Erratum: Parameterized quantum circuits as machine learning models (2019 Quant. Sci. Tech. 4 043001)," *Quantum Sci. Technol.*, vol. 5, no. 1, Dec. 2019, Art. no. 019601.

[7] K. Beer, D. Bondarenko, T. Farrelly, T. J. Osborne, R. Salzmann, D. Scheiermann, and R. Wolf, "Training deep quantum neural networks," *Nature Commun.*, vol. 11, no. 1, p. 808, Feb. 2020.

[8] P. Lamichhane and D. B. Rawat, "Quantum machine learning: Recent advances, challenges, and perspectives," *IEEE Access*, vol. 13, pp. 94057–94105, 2025.

[9] Y. Wang and J. Liu, "A comprehensive review of quantum machine learning: From NISQ to fault tolerance," *Rep. Prog. Phys.*, vol. 87, no. 11, Nov. 2024, Art. no. 116402.

[10] D. Peral-García, J. Cruz-Benito, and F. J. García-Peñalvo, "Systematic literature review: Quantum machine learning and its applications," *Comput. Sci. Rev.*, vol. 51, Feb. 2024, Art. no. 100619.

[11] J. C. L. Chow, "Quantum computing in medicine," *Med. Sci.*, vol. 12, no. 4, p. 67, Nov. 2024.

[12] J. C. L. Chow, "Quantum computing and machine learning in medical decision-making: A comprehensive review," *Algorithms*, vol. 18, no. 3, p. 156, Mar. 2025.

[13] W. K. Wootters and W. H. Zurek, "A single quantum cannot be cloned," *Nature*, vol. 299, no. 5886, pp. 802–803, Oct. 1982.

[14] M. A. Khan, M. N. Aman, and B. Sikdar, "Beyond bits: A review of quantum embedding techniques for efficient information processing," *IEEE Access*, vol. 12, pp. 46118–46137, 2024.

[15] S. Mourya, B. R. L. Cour, and B. D. Sahoo, "Emulation of quantum algorithms using CMOS analog circuits," *IEEE Trans. Quantum Eng.*, vol. 4, pp. 1–16, 2023.

[16] A. King et al., "Beyond-classical computation in quantum simulation," *Science*, vol. 388, no. 6743, pp. 199–204, 2025.

[17] A. Ceselli and M. Premoli, "On good encodings for quantum annealer and digital optimization solvers," *Sci. Rep.*, vol. 13, no. 1, Apr. 2023, Art. no. 1.

[18] M. D'Arcangelo, L.-P. Henry, L. Henriet, D. Loco, N. Gouraud, S. Angebault, J. Sueiro, J. Forêt, P. Monmarché, and J.-P. Piquemal, "Leveraging analog quantum computing with neutral atoms for solvent configuration prediction in drug discovery," *Phys. Rev. Res.*, vol. 6, no. 4, Oct. 2024, Art. no. 043020.

[19] V. Kendon, K. Nemoto, and W. J. Munro, "Quantum analogue computing," *Phil. Trans. Roy. Soc. A, Math.*, vol. 368, no. 1924, pp. 3609–3620, 2010.

[20] J. Wurtz, A. Bylinskii, B. Braverman, J. Amato-Grill, S. H. Cantu, F. Huber, A. Lukin, F. Liu, P. Weinberg, J. Long, S.-T. Wang, N. Gemelke, and A. Keesling, "Aquila: QuEra's 256-qubit neutral-atom quantum computer," 2023, *arXiv:2306.11727*.

[21] T. Kehrer, T. Nadolny, and C. Bruder, "Improving transmon qudit measurement on IBM quantum hardware," *Phys. Rev. Res.*, vol. 6, no. 1, Jan. 2024, Art. no. 013050.

[22] H. K. Warner, J. Holzgrafe, B. Yankelevich, D. Barton, S. Poletto, C. J. Xin, N. Sinclair, D. Zhu, E. Sete, B. Langley, E. Batson, M. Colangelo, A. Shams-Ansari, G. Joe, K. K. Berggren, L. Jiang, M. J. Reagor, and M. Lončar, "Coherent control of a superconducting qubit using light," *Nature Phys.*, vol. 21, no. 5, pp. 831–838, Apr. 2025, doi: 10.1038/s41567-025-02812-0.

[23] S. Debnath, N. M. Linke, C. Figgatt, K. A. Landsman, K. Wright, and C. Monroe, "Demonstration of a small programmable quantum computer with atomic qubits," *Nature*, vol. 536, no. 7614, pp. 63–66, Aug. 2016.

[24] N. Killoran, J. Izaac, N. Quesada, V. Bergholm, M. Amy, and C. Weedbrook, "Strawberry fields: A software platform for photonic quantum computing," *Quantum*, vol. 3, p. 129, Mar. 2019.

[25] P. Murali, N. M. Linke, M. Martonosi, A. J. Abhari, N. H. Nguyen, and C. H. Alderete, "Full-stack, real-system quantum computer studies: Architectural comparisons and design insights," in *Proc. ACM/IEEE 46th Annu. Int. Symp. Comput. Archit. (ISCA)*, Jun. 2019, pp. 527–540.

[26] IBM Quantum Documentation. *IBM Quantum Documentation*. Accessed: May 3, 2025. [Online]. Available: https://docs.quantum.ibm.com/

[27] InfoQ. *Google Cirq: A Python Open Source Library for Quantum Computing*. Accessed: Mar. 26, 2024. [Online]. Available: https://www.infoq.com/news/2018/07/google-cirq-quantum-library/

[28] J. R. Johansson, P. D. Nation, and F. Nori, "QuTiP 2: A Python framework for the dynamics of open quantum systems," *Comput. Phys. Commun.*, vol. 184, no. 4, pp. 1234–1240, Apr. 2013.

[29] *PennyLane Documentation—PennyLane 0.34.0 Documentation*. Accessed: Jan. 24, 2024. [Online]. Available: https://docs.pennylane.ai/en/stable/

[30] D. S. Steiger, T. Häner, and M. Troyer, "ProjectQ: An open source software framework for quantum computing," *Quantum*, vol. 2, p. 49, Jan. 2018.

[31] J.-S. Kim, A. McCaskey, B. Heim, M. Modani, S. Stanwyck, and T. Costa, "CUDA quantum: The platform for integrated quantum-classical computing," in *Proc. 60th ACM/IEEE Design Autom. Conf. (DAC)*, Jul. 2023, pp. 1–4.

[32] H. Bayraktar et al., "CuQuantum SDK: A high-performance library for accelerating quantum science," in *Proc. IEEE Int. Conf. Quantum Comput. Eng. (QCE)*, Sep. 2023, pp. 1050–1061.

[33] E. Gil-Fuster, J. Eisert, and C. Bravo-Prieto, "Understanding quantum machine learning also requires rethinking generalization," *Nature Commun.*, vol. 15, no. 1, p. 2277, Mar. 2024.

[34] M. Schuld and F. Petruccione, *Supervised Learning With Quantum Computers*. Cham, Switzerland: Springer, 2018.

[35] H. Yano, Y. Suzuki, K. M. Itoh, R. Raymond, and N. Yamamoto, "Efficient discrete feature encoding for variational quantum classifier," *IEEE Trans. Quantum Eng.*, vol. 2, pp. 1–14, 2021.

[36] S. Thanasilp, S. Wang, M. Cerezo, and Z. Holmes, "Exponential concentration in quantum kernel methods," *Nature Commun.*, vol. 15, no. 1, p. 5200, Jun. 2024.

[37] X. Zhou, J. Yu, J. Tan, and T. Jiang, "Quantum kernel estimation-based quantum support vector regression," *Quantum Inf. Process.*, vol. 23, no. 1, p. 29, Jan. 2024.

[38] X. Vasques, H. Paik, and L. Cif, "Application of quantum machine learning using quantum kernel algorithms on multiclass neuron M-type classification," *Sci. Rep.*, vol. 13, no. 1, p. 11541, Jul. 2023.

[39] R. Zhang, J. Wang, N. Jiang, and Z. Wang, "Quantum support vector machine without iteration," *Inf. Sci.*, vol. 635, pp. 25–41, Jul. 2023.

[40] M. Incudini, M. Grossi, A. Mandarino, S. Vallecorsa, A. Di Pierro, and D. Windridge, "The quantum path kernel: A generalized neural tangent kernel for deep quantum machine learning," *IEEE Trans. Quantum Eng.*, vol. 4, pp. 1–17, 2023.

[41] Y. Du, Y. Qian, X. Wu, and D. Tao, "A distributed learning scheme for variational quantum algorithms," *IEEE Trans. Quantum Eng.*, vol. 3, pp. 1–16, 2022.

[42] M. Simonetti, D. Perri, and O. Gervasi, "Variational methods in optical quantum machine learning," *IEEE Access*, vol. 11, pp. 131394–131408, 2023.

[43] J. H. A. de Carvalho and F. M. de Paula Neto, "Parametrized constant-depth quantum neuron," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 15932–15943, Nov. 2024.

[44] S. Park and O. Simeone, "Quantum conformal prediction for reliable uncertainty quantification in quantum machine learning," *IEEE Trans. Quantum Eng.*, vol. 5, pp. 1–24, 2023.

[45] Z. Wang, M. Xu, and Y. Zhang, "Quantum pulse coupled neural network," *Neural Netw.*, vol. 152, pp. 105–117, Jan. 2022.

[46] N. Nguyen and K.-C. Chen, "Bayesian quantum neural networks," *IEEE Access*, vol. 10, pp. 54110–54122, 2022.

[47] Y. Chen, "Quantum dilated convolutional neural networks," *IEEE Access*, vol. 10, pp. 20240–20246, 2022.

[48] A. Mari, T. R. Bromley, J. Izaac, M. Schuld, and N. Killoran, "Transfer learning in hybrid classical-quantum neural networks," *Quantum*, vol. 4, p. 340, Oct. 2020.

[49] E. Zardini, A. Delilbasic, E. Blanzieri, G. Cavallaro, and D. Pastorello, "Local binary and multiclass SVMs trained on a quantum annealer," *IEEE Trans. Quantum Eng.*, vol. 5, pp. 1–13, 2024.

[50] A. Kottahachchi Kankanamge Don, I. Khalil, and M. Atiquzzaman, "A fusion of supervised contrastive learning and variational quantum classifiers," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 770–779, Feb. 2024.

[51] R. Huang, X. Tan, and Q. Xu, "Quantum federated learning with decentralized data," *IEEE J. Sel. Topics Quantum Electron.*, vol. 28, no. 4, pp. 1–10, Jul. 2022.

[52] N. V. Nguyen and K. Chen, "Quantum embedding search for quantum machine learning," *IEEE Access*, vol. 10, pp. 41444–41456, 2022.

[53] N. Singh, M. Kumar, and B. Biswas, "PQCLP: Parameterized quantum circuit based link prediction in dynamic networks," *Comput. Netw.*, vol. 241, Mar. 2023, Art. no. 110210.

[54] D. Maheshwari, D. Sierra-Sosa, and B. Garcia-Zapirain, "Variational quantum classifier for binary classification: Real vs synthetic dataset," *IEEE Access*, vol. 10, pp. 3705–3715, 2022.

[55] M. Aly, S. Fadaaq, O. A. Warga, Q. Nasir, and M. A. Talib, "Experimental benchmarking of quantum machine learning classifiers," in *Proc. 6th Int. Conf. Signal Processing Inf. Security (ICSPIS)*, Dubai, UAE, Nov. 2023, pp. 240–245.

[56] N. M. De Oliveira, D. K. Park, I. F. Araujo, and A. J. Da Silva, "Quantum variational distance-based centroid classifier," *Neurocomputing*, vol. 576, Apr. 2024, Art. no. 127356.

[57] P. Singh, "FQTSFM: A fuzzy-quantum time series forecasting model," *Inf. Sci.*, vol. 566, pp. 57–79, Jan. 2021.

[58] I. Kerenidis, J. Landman, A. Luongo, and A. Prakash, "Q-means: A quantum algorithm for unsupervised machine learning," in *Proc. Adv. Neural Inf. Process. Systems*, 2019, pp. 1–11.

[59] R. K. Singh, S. K. Meher, and S. K. Rout, "New single-preparation methods for unsupervised quantum machine learning problems," *IEEE Access*, vol. 9, pp. 130257–130270, 2021.

[60] A. Yadav, S. Roy, P. Mishra, A. Tripathi, and P. Tiwari, "Quantum model regression for generating fuzzy numbers in adiabatic quantum computing," *Phys. A, Stat. Mech. Appl.*, vol. 639, Feb. 2024, Art. no. 129699.

[61] Z. Ye, K. Yu, G.-D. Guo, and S. Lin, "Quantum self-organizing feature mapping neural network algorithm based on Grover search algorithm," *Phys. A, Stat. Mech. Appl.*, vol. 639, Apr. 2024, Art. no. 129690.

[62] D. Dong and C. Chen, "Quantum reinforcement learning," *IEEE Trans. Syst., Man, Cybern., B, Cybern.*, vol. 38, no. 5, pp. 1207–1220, May 2008.

[63] Z. He, Y. Zhao, L. Jiang, and Y.-A. Yan, "Curriculum-based deep reinforcement learning for quantum control," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8852–8865, Nov. 2023.

[64] R. Ravish, N. R. Bhat, N. Nandakumar, S. Sagar, Sunil, and P. B. Honnavalli, "Optimization of reinforcement learning using quantum computation," *IEEE Access*, vol. 12, pp. 179396–179417, 2024.

[65] B. Pang, Y. Qin, Y. Liu, Y. Li, C. Xiang, and Y. Liu, "Quantum imitation learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 14190–14204, Oct. 2024.

[66] H. Hohenfeld, D. Heimann, F. Wiebe, and F. Kirchner, "Quantum deep reinforcement learning for robot navigation tasks," *IEEE Access*, vol. 12, pp. 87217–87236, 2024.

[67] R. Yan, Y. Wang, Y. Xu, and J. Dai, "A multiagent quantum deep reinforcement learning method for distributed frequency control of islanded microgrids," *IEEE Trans. Control Netw. Syst.*, vol. 9, no. 4, pp. 1622–1632, Dec. 2022.
[68] S. Chaudhary, I. Budhiraja, R. Chaudhary, N. Kumar, D. Garg, and A. M. Almuhaideb, "Quantum federated reinforcement-learning-based joint mode selection and resource allocation for STAR-RIS-aided VRCS," *IEEE Internet Things J.*, vol. 11, no. 22, pp. 36242–36256, Nov. 2024.
[69] K. Reuer, J. Landgraf, T. Fösel, J. O'Sullivan, L. Beltrán, A. Akin, G. J. Norris, A. Remm, M. Kerschbaum, J.-C. Besse, F. Marquardt, A. Wallraff, and C. Eichler, "Realizing a deep reinforcement learning agent for real-time quantum feedback," *Nature Commun.*, vol. 14, no. 1, p. 7138, Nov. 2023.
[70] L. Yin, L. Chen, D. Liu, X. Huang, and F. Gao, "Quantum deep reinforcement learning for rotor side converter control of double-fed induction generator-based wind turbines," *Eng. Appl. Artif. Intell.*, vol. 106, Nov. 2021, Art. no. 104451.
[71] K. Nakaji and N. Yamamoto, "Quantum semi-supervised generative adversarial network for enhanced data classification," *Sci. Rep.*, vol. 11, no. 1, p. 18932, Oct. 2021.
[72] S. Saeedi, A. Panahi, and T. Arodz, "Quantum semi-supervised kernel learning," *Quantum Mach. Intell.*, vol. 3, no. 2, p. 5, Dec. 2021.
[73] S. Yarkoni, A. Kleshchonok, Y. Dzerin, F. Neukart, and M. Hilbert, "Semi-supervised time series classification method for quantum computing," *Quantum Mach. Intell.*, vol. 3, no. 1, p. 2, Jun. 2021.
[74] R.-X. Zhao, J. Shi, and X. Li, "QKSAN: A quantum kernel self-attention network," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10184–10195, Dec. 2024.
[75] I. Cong, S. Choi, and M. D. Lukin, "Quantum convolutional neural networks," *Nature Phys.*, vol. 15, no. 12, pp. 1273–1278, Dec. 2019.
[76] M. Henderson, S. Shakya, S. Pradhan, and T. Cook, "Quanvolutional neural networks: Powering image recognition with quantum circuits," 2019, *arXiv:1904.04767*.
[77] P.-L. Dallaire-Demers and N. Killoran, "Quantum generative adversarial networks," *Phys. Rev. A*, vol. 98, no. 1, Jul. 2018, Art. no. 012324.
[78] C. Zoufal, A. Lucchi, and S. Woerner, "Quantum generative adversarial networks for learning and loading random distributions," *npj Quantum Inf.*, vol. 5, no. 1, p. 103, Nov. 2019.
[79] Y. Li, Z. Wang, R. Han, S. Shi, J. Li, R. Shang, H. Zheng, G. Zhong, and Y. Gu, "Quantum recurrent neural networks for sequential learning," *Neural Netw.*, vol. 166, pp. 148–161, Sep. 2023.
[80] S. Y. Chen, S. Yoo, and Y. L. Fang, "Quantum long short-term memory," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 8622–8626.
[81] A. Marcianò, D. Chen, F. Fabrocini, C. Fields, E. Greco, N. Gresnigt, K. Jinklub, M. Lulli, K. Terzidis, and E. Zappala, "Quantum neural networks and topological quantum field theories," *Neural Netw.*, vol. 153, pp. 164–178, Sep. 2022.
[82] J. Zheng, Q. Gao, M. Ogorzałek, J. Lü, and Y. Deng, "A quantum spatial graph convolutional neural network model on quantum circuits," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 3, pp. 5706–5720, Mar. 2025.
[83] S. R. Vadyala and S. N. Betgeri, "General implementation of quantum physics-informed neural networks," *Array*, vol. 18, Jul. 2023, Art. no. 100287.
[84] E. Ovalle-Magallanes, D. E. Alvarado-Carrillo, J. G. Avina-Cervantes, I. Cruz-Aceves, and J. Ruiz-Pinales, "Quantum angle encoding with learnable rotation applied to quantum–classical convolutional neural networks," *Appl. Soft Comput.*, vol. 141, Jul. 2023, Art. no. 110307.
[85] Q.-W. Zeng, H.-Y. Ge, C. Gong, and N.-R. Zhou, "Conditional quantum circuit born machine based on a hybrid quantum–classical framework," *Phys. A, Stat. Mech. Appl.*, vol. 618, May 2023, Art. no. 128693.
[86] L. Liu, T. Song, Z. Sun, and J. Lei, "Quantum generative adversarial networks based on Rényi divergences," *Phys. A, Stat. Mech. Appl.*, vol. 607, Dec. 2022, Art. no. 128169.
[87] A. O'Dwyer Boyle and R. Nikandish, "A hybrid quantum-classical generative adversarial network for near-term quantum processors," *IEEE Access*, vol. 12, pp. 102688–102701, 2024.
[88] L. Bischof, S. Teodoropol, R. M. Füchslin, and K. Stockinger, "Hybrid quantum neural networks show strongly reduced need for free parameters in entity matching," *Sci. Rep.*, vol. 15, no. 1, p. 4318, Feb. 2025.
[89] S. Y.-C. Chen, C.-M. Huang, C.-W. Hsing, and Y.-J. Kao, "An end-to-end trainable hybrid classical-quantum classifier," *Mach. Learning: Sci. Technol.*, vol. 2, no. 4, Sep. 2021, Art. no. 045021.
[90] S. Kumar, S. Dangwal, and D. Bhowmik, "Supervised learning using a dressed quantum network with 'super compressed encoding': Algorithm and quantum-hardware-based implementation," 2020, *arXiv:2007.10242*.
[91] S. Wu, Y. Zhang, and J. Li, "Quantum data parallelism in quantum neural networks," *Phys. Rev. Res.*, vol. 7, no. 1, Feb. 2025, Art. no. 013177.
[92] L. Kim, S. Lloyd, and M. Marvian, "Hamiltonian quantum generative adversarial networks," *Phys. Rev. Res.*, vol. 6, no. 3, Jul. 2024, Art. no. 033019.
[93] T. A. Ngo, T. Nguyen, and T. C. Thang, "A survey of recent advances in quantum generative adversarial networks," *Electronics*, vol. 12, no. 4, p. 856, Feb. 2023.
[94] J. Li, R. O. Topaloglu, and S. Ghosh, "Quantum generative models for small molecule drug discovery," *IEEE Trans. Quantum Eng.*, vol. 2, pp. 1–8, 2021.
[95] Z. Qu, W. Shi, and P. Tiwari, "Quantum conditional generative adversarial network based on patch method for abnormal electrocardiogram generation," *Comput. Biol. Med.*, vol. 166, Nov. 2023, Art. no. 107549.
[96] J. Bausch, "Recurrent quantum neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 1368–1379.
[97] F. Chen, Q. Zhao, L. Feng, C. Chen, Y. Lin, and J. Lin, "Quantum mixed-state self-attention network," *Neural Netw.*, vol. 185, May 2025, Art. no. 107123.
[98] G. Li, X. Zhao, and X. Wang, "Quantum self-attention neural networks for text classification," *Sci. China Inf. Sci.*, vol. 67, no. 4, Mar. 2024, Art. no. 142501.
[99] E. A. Cherrat, I. Kerenidis, N. Mathur, J. Landman, M. Strahm, and Y. Y. Li, "Quantum vision transformers," *Quantum*, vol. 8, p. 1265, Feb. 2024.
[100] J. Lu, Y. Xu, Y. Zhao, J. Wu, and J. Zhu, "Quantum-classical image processing for scene classification," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 8, pp. 5282–5290, Aug. 2022.
[101] Y. Wang, Y. Wang, C. Chen, R. Jiang, and W. Huang, "Development of variational quantum deep neural networks for image recognition," *Neurocomputing*, vol. 501, pp. 566–582, Aug. 2022.
[102] I. Ahmad, A. Mohammed, I. Hussain, M. Khan, A. Irtaza, and A. Rehman, "A robust solution for recognizing accurate handwritten text extraction using quantum convolutional neural network and transformer models," *Comput. Electr. Eng.*, vol. 114, Jul. 2024, Art. no. 108564.
[103] Z. Chen, J. Ma, W. Zeng, and Z. Zhang, "Hybrid quantum-classical generative adversarial networks for image generation via learning discrete distribution," *Future Gener. Comput. Syst.*, vol. 135, pp. 47–55, Dec. 2022.
[104] S. L. Tsang, M. T. West, S. M. Erfani and M. Usman, "Hybrid quantum–classical generative adversarial network for high-resolution image generation," *IEEE Trans. Quantum Eng.*, vol. 4, pp. 1–19, Dec. 2022.
[105] X. Zhao, F. He, X. Guo, and F. Sun, "Efficient quantum image classification using single qubit encoding," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 6, no. 5, pp. 965–975, May 2022.
[106] Y. Zhou, L. Yu, Y. Liu, and W. Sun, "QE-DAL: A quantum image feature extraction with dense distribution-aware learning framework for object counting and localization," *Appl. Soft Comput.*, vol. 136, Jan. 2023, Art. no. 110160.
[107] Y. Zhu, A. Bouridane, M. E. Celebi, D. Konar, P. Angelov, Q. Ni, and R. Jiang, "Quantum face recognition with multigate quantum convolutional neural network," *IEEE Trans. Artif. Intell.*, vol. 5, no. 12, pp. 6330–6341, Dec. 2024.
[108] W. Zhai, X. Xing, and G. Jeon, "Region-aware quantum network for crowd counting," *IEEE Trans. Consumer Electronics*, vol. 70, no. 3, pp. 5536–5544, Aug. 2024.
[109] L.-H. Gong, J.-J. Pei, T.-F. Zhang, and N.-R. Zhou, "Quantum convolutional neural network based on variational quantum circuits," *Opt. Commun.*, vol. 550, Jan. 2024, Art. no. 129993.
[110] T. A. Akinola, X. Li, R. Wilkins, P. H. Obiomon, and L. Qian, "Robust inverse quantum fourier transform inspired algorithm for unsupervised image segmentation," *IEEE Access*, vol. 12, pp. 99029–99044, Jan. 2024.
[111] S. Altares-López, J. J. García-Ripoll, and A. Ribeiro, "AutoQML: Automatic generation and training of robust quantum-inspired classifiers by using evolutionary algorithms on grayscale images," *Expert Syst. Appl.*, vol. 244, Jun. 2024, Art. no. 122984.
[112] L.-H. Yu, X.-Y. Li, G. Chen, Q.-S. Zhu, H. Li, and G.-W. Yang, "QUSL: Quantum unsupervised image similarity learning with enhanced performance," *Expert Syst. Appl.*, vol. 258, Dec. 2024, Art. no. 125112.
[113] M. Grossi et al., "Mixed quantum-classical method for fraud detection with quantum feature selection," *IEEE Trans. Quantum Eng.*, vol. 3, pp. 1–12, Jan. 2022.

[114] M. S. Salek, P. K. Biswas, J. Pollard, J. Hales, Z. Shen, V. Dixit, M. Chowdhury, S. M. Khan, and Y. Wang, "A novel hybrid quantum-classical framework for an in-vehicle controller area network intrusion detection," *IEEE Access*, vol. 11, pp. 96081–96092, 2023.
[115] A. Tudisco, D. Volpe, G. Ranieri, G. Curato, D. Ricossa, M. Graziano, and D. Corbelletto, "Evaluating the computational advantages of the variational quantum circuit model in financial fraud detection," *IEEE Access*, vol. 12, pp. 102918–102940, 2024.
[116] C. Huot, S. Heng, T.-K. Kim, and Y. Han, "Quantum autoencoder for enhanced fraud detection in imbalanced credit card dataset," *IEEE Access*, vol. 12, pp. 169671–169682, 2024.
[117] M. Zielewski, K. Takahashi, Y. Shimomura, and H. Takizawa, "Efficient pause location prediction using quantum annealing simulations and machine learning," *IEEE Access*, vol. 11, pp. 104285–104294, Oct. 2023.
[118] Y.-Y. Hong, C. J. E. Arce, and T.-W. Huang, "A robust hybrid classical and quantum model for short-term wind speed forecasting," *IEEE Access*, vol. 11, pp. 90811–90824, 2023.
[119] Y. Y. Hong and J. B. D. Santos, "Day-ahead spatiotemporal wind speed forecasting based on a hybrid model of quantum and residual long short-term memory optimized by particle swarm algorithm," *IEEE Syst. J.*, vol. 17, no. 4, pp. 6081–6092, Dec. 2023.
[120] Y. Yu, G. Hu, C. Liu, J. Xiong, and Z. Wu, "Prediction of solar irradiance one hour ahead based on quantum long short-term memory network," *IEEE Trans. Quantum Eng.*, vol. 4, pp. 1–15, Jan. 2023.
[121] Y.-Y. Hong, D. Josh Domingo Lopez, and Y.-Y. Wang, "Solar irradiance forecasting using a hybrid quantum neural network: A comparison on GPU-based workflow development platforms," *IEEE Access*, vol. 12, pp. 145079–145094, 2024.
[122] J. Kumar, D. Saxena, A. K. Singh, and A. V. Vasilakos, "A quantum controlled-NOT neural network-based load forecast and management model for smart grid," *IEEE Syst. J.*, vol. 17, no. 4, pp. 5714–5725, Dec. 2023.
[123] S.-G. Jeong, Q. V. Do, and W.-J. Hwang, "Short-term photovoltaic power forecasting based on hybrid quantum gated recurrent unit," *ICT Exp.*, vol. 10, no. 3, pp. 608–613, Jun. 2024.
[124] Z. Qu, W. Chen, and P. Tiwari, "HQ-DCGAN: Hybrid quantum deep convolutional generative adversarial network approach for ECG generation," *Knowledge-Based Syst.*, vol. 301, Oct. 2024, Art. no. 112260.
[125] K. El amraoui, Z. Pu, L. Koutti, L. Masmoudi, and J. Valente de Oliveira, "A super resolution method based on generative adversarial networks with quantum feature enhancement: Application to aerial agricultural images," *Neurocomputing*, vol. 577, Apr. 2024, Art. no. 127346.
[126] S. Park, H. Baek, and J. Kim, "Spatio-temporal multi-metaverse dynamic streaming for hybrid quantum-classical systems," *IEEE/ACM Trans. Netw.*, vol. 32, no. 6, pp. 5279–5294, Dec. 2024.
[127] M. Emu, S. Choudhury, and K. Salomaa, "Stochastic resource optimization for metaverse data marketplace by leveraging quantum neural networks," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 3, pp. 2613–2623, Jun. 2024.
[128] C. Ren, Z. Y. Dong, H. Yu, M. Xu, Z. Xiong, and D. Niyato, "ESQFL: Digital twin-driven explainable and secured quantum federated learning for voltage stability assessment in smart grids," *IEEE J. Sel. Topics Signal Process.*, vol. 18, no. 5, pp. 964–978, Jul. 2024.
[129] Silvirianti, B. Narottama, and S. Y. Shin, "Layerwise quantum deep reinforcement learning for joint optimization of UAV trajectory and resource allocation," *IEEE Internet Things J.*, vol. 11, no. 1, pp. 430–443, Jan. 2024.
[130] F. Yousef Alghayadh, J. Venkata Naga Ramesh, I. Keshta, M. Soni, R. Rivera, K. D. V. Prasad, A. Muhammad Soomar, and M. Tiwari, "Quantum target recognition enhancement algorithm for UAV consumer applications," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 5553–5560, Aug. 2024.
[131] I. Naz, J. H. Shah, M. H. U. Rehman, M. Rafiq, and G. S. Choi, "Quantum mechanism-based convolution model for the classification of pathogenic bacteria," *IEEE Access*, vol. 11, pp. 137747–137757, 2023.
[132] M. Gandhudi, P. J. A. Alphonse, U. Fiore, and G. R. Gangadharan, "Explainable hybrid quantum neural networks for analyzing the influence of tweets on stock price prediction," *Comput. Electr. Eng.*, vol. 118, Aug. 2024, Art. no. 109302.
[133] W. Yu, L. Yin, C. Zhang, Y. Chen, and A. X. Liu, "Application of quantum recurrent neural network in low-resource language text classification," *IEEE Trans. Quantum Eng.*, vol. 5, pp. 1–13, 2024.
[134] A. Phukan, S. Pal, and A. Ekbal, "Hybrid quantum-classical neural network for multimodal multitask sarcasm, emotion, and sentiment analysis," *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 5, pp. 5740–5750, Oct. 2024.
[135] P. Tiwari, L. Zhang, Z. Qu, and G. Muhammad, "Quantum fuzzy neural network for multimodal sentiment and sarcasm detection," *Inf. Fusion*, vol. 103, Mar. 2024, Art. no. 102085.
[136] A. Omar and T. Abd El-Hafeez, "Quantum computing and machine learning for Arabic language sentiment classification in social media," *Sci. Rep.*, vol. 13, no. 1, p. 17305, Oct. 2023.
[137] S. Ruan, Z. Liang, Q. Guan, P. Griffin, X. Wen, Y. Lin, and Y. Wang, "VIOLET: Visual analytics for explainable quantum neural networks," *IEEE Trans. Vis. Comput. Graph.*, pp. 2862–2874, Jun. 2023.
[138] U. Ullah, A. G. O. Jurado, I. D. Gonzalez, and B. Garcia-Zapirain, "A fully connected quantum convolutional neural network for classifying ischemic cardiopathy," *IEEE Access*, vol. 10, pp. 134592–134605, 2022.
[139] S. Prabhu, S. Gupta, G. M. Prabhu, A. V. Dhanuka and K. V. Bhat, "QuCardio: Application of quantum machine learning for detection of cardiovascular diseases," *IEEE Access*, vol. 11, pp. 136122–136135, Jan. 2023.
[140] D. Konar, S. Bhattacharyya, B. K. Panigrahi, and E. C. Behrman, "Qutrit-inspired fully self-supervised shallow quantum learning network for brain tumor segmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6331–6345, Nov. 2022.
[141] C. Olvera, O. M. Ross, and Y. Rubio, "EEG-based motor imagery classification with quantum algorithms," *Expert Syst. Appl.*, vol. 247, Aug. 2024, Art. no. 123354.
[142] J. Y. Yu, W. S. Sim, J. Y. Jung, S. H. Park, H. S. Kim, and Y. R. Park, "Evaluation of conventional and quantum computing for predicting mortality based on small early-onset colorectal cancer data," *Appl. Soft Comput.*, vol. 162, Sep. 2024, Art. no. 111781.
[143] G. Swathi, A. Altalbe, and R. P. Kumar, "QuCNet: Quantum-inspired convolutional neural networks for optimized thyroid nodule classification," *IEEE Access*, vol. 12, pp. 27829–27842, 2024.
[144] K. K. Makam, V. K. Singh, and R. B. Pachori, "ALS detection framework based on automatic singular spectrum analysis and quantum convolutional neural network from EMG signals," *IEEE Sensors Lett.*, vol. 8, no. 9, pp. 1–4, Sep. 2024.
[145] A. Jenber Belay, Y. M. Walle, and M. B. Haile, "Deep ensemble learning and quantum machine learning approach for Alzheimer's disease detection," *Sci. Rep.*, vol. 14, no. 1, p. 14196, Jun. 2024.
[146] E. Ovalle-Magallanes, J. G. Avina-Cervantes, I. Cruz-Aceves, and J. Ruiz-Pinales, "Hybrid classical–quantum convolutional neural network for stenosis detection in X-ray coronary angiography," *Expert Syst. Appl.*, vol. 189, Mar. 2022, Art. no. 116112.
[147] M. Yousif, B. Al-Khateeb, and B. Garcia-Zapirain, "A new quantum circuits of quantum convolutional neural network for X-ray images classification," *IEEE Access*, vol. 12, pp. 65660–65671, 2024.
[148] V. M. R. Sankari, U. Umapathy, S. Alasmari, and S. M. Aslam, "Automated detection of retinopathy of prematurity using quantum machine learning and deep learning techniques," *IEEE Access*, vol. 11, pp. 94306–94321, 2023.
[149] R. K. Ahalya, U. Snekhalatha, and V. Dhanraj, "Automated segmentation and classification of hand thermal images in rheumatoid arthritis using machine learning algorithms: A comparison with quantum machine learning technique," *J. Thermal Biol.*, vol. 111, Jan. 2023, Art. no. 103404.
[150] S. R. Gadu and C. S. Potala, "A hybrid generative adversarial network with quantum U-NET for 3D spine X-ray image registration," *Healthcare Analytics*, vol. 4, Dec. 2023, Art. no. 100251.
[151] S. S. Reka, H. L. Karthikeyan, A. J. Shakil, P. Venugopal, and M. Muniraj, "Exploring quantum machine learning for enhanced skin lesion classification: A comparative study of implementation methods," *IEEE Access*, vol. 12, pp. 104568–104584, 2024.
[152] E. Paquet, F. Soleymani, G. St-Pierre-Lemieux, H. L. Viktor, and W. Michalowski, "QuantumBound – interactive protein generation with one-shot learning and hybrid quantum neural networks," *Artif. Intell. Chem.*, vol. 2, no. 1, Jun. 2024, Art. no. 100030.
[153] A. S. Bhatia, M. K. Saggi, and S. Kais, "Quantum machine learning predicting ADME-tox properties in drug discovery," *J. Chem. Inf. Model.*, vol. 63, no. 21, pp. 6476–6486, Nov. 2023.
[154] D. A. Shoieb, A. Younes, S. M. Youssef, and K. M. Fathalla, "HQMC-CPC: A hybrid quantum multiclass cardiac pathologies classification integrating a modified hardware efficient ansatz," *IEEE Access*, vol. 12, pp. 18295–18314, 2024.

[155] S. Hossain, S. Umer, R. K. Rout, and H. A. Marzouqi, "A deep quantum convolutional neural network based facial expression recognition for mental health analysis," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 32, pp. 1556–1565, 2024.
[156] A. Padha and A. Sahoo, "QCLR: Quantum-LSTM contrastive learning framework for continuous mental health monitoring," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 121921.
[157] K. Yu, C. Chakraborty, D. Xu, T. Zhang, H. Zhu, O. Alfarraj, and A. Tolba, "Hybrid quantum classical optimization for low-carbon sustainable edge architecture in RIS-assisted AIoT healthcare systems," *IEEE Internet Things J.*, vol. 11, no. 24, pp. 38987–38998, Dec. 2024.
[158] J. Zhang, J. Tian, M. Li, J. I. Leon, L. G. Franquelo, H. Luo, and S. Yin, "A parallel hybrid neural network with integration of spatial and temporal features for remaining useful life prediction in prognostics," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.
[159] C. B. S. Maior, L. M. M. Araújo, I. D. Lins, M. D. C. Moura, and E. L. Droguett, "Prognostics and health management of rotating machinery via quantum machine learning," *IEEE Access*, vol. 11, pp. 25132–25151, 2023.
[160] Z. Qu, Y. Li, and P. Tiwari, "QNMF: A quantum neural network based multimodal fusion system for intelligent diagnosis," *Inf. Fusion*, vol. 100, Dec. 2023, Art. no. 101913.
[161] Z. Qu, W. Shi, B. Liu, D. Gupta, and P. Tiwari, "IoMT-based smart healthcare detection system driven by quantum blockchain and quantum neural network," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 6, pp. 3317–3328, Jun. 2024.
[162] E. Hassan, M. S. Hossain, A. Saber, S. Elmougy, A. Ghoneim, and G. Muhammad, "A quantum convolutional network and ResNet (50)-based classification architecture for the MNIST medical dataset," *Biomed. Signal Process. Control*, vol. 87, Jan. 2024, Art. no. 105560.
[163] B. Narottama and S. Y. Shin, "Federated quantum neural network with quantum teleportation for resource optimization in future wireless communication," *IEEE Trans. Veh. Technol.*, vol. 72, no. 11, pp. 14717–14733, Nov. 2023.
[164] J. A. Ansere, E. Gyamfi, V. Sharma, H. Shin, O. A. Dobre, and T. Q. Duong, "Quantum deep reinforcement learning for dynamic resource allocation in mobile edge computing-based IoT systems," *IEEE Trans. Wireless Commun.*, vol. 23, no. 6, pp. 6221–6233, Jun. 2024.
[165] J. Adu Ansere, D. T. Tran, O. A. Dobre, H. Shin, G. K. Karagiannidis, and T. Q. Duong, "Energy-efficient optimization for mobile edge computing with quantum machine learning," *IEEE Wireless Commun. Lett.*, vol. 13, no. 3, pp. 661–665, Mar. 2024.
[166] B. Narottama and S. Y. Shin, "Quantum neural networks for resource allocation in wireless communications," *IEEE Trans. Wireless Commun.*, vol. 21, no. 2, pp. 1103–1116, Feb. 2022.
[167] J. Gao, J. Wang, J. Gu, and W. Shi, "Quantum-based deep Q-network bandwidth resource allocation algorithm for UASN," *IEEE Internet Things J.*, vol. 11, no. 24, pp. 39932–39940, Dec. 2024.
[168] I. Gupta, D. Saxena, A. K. Singh, and C.-N. Lee, "A multiple controlled Toffoli driven adaptive quantum neural network model for dynamic workload prediction in cloud environments," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 7574–7588, Dec. 2024.
[169] H. Yang, X. Li, Z. Liu, and W. Pedrycz, "Improved differential privacy noise mechanism in quantum machine learning," *IEEE Access*, vol. 11, pp. 50157–50164, 2023.
[170] R. W. Wardhani, D. S. C. Putranto, T.-T.-H. Le, J. Ji, and H. Kim, "Toward hybrid classical deep learning-quantum methods for steganalysis," *IEEE Access*, vol. 12, pp. 45238–45252, 2024.
[171] R. Gupta, D. Saxena, I. Gupta, A. Makkar, and A. Kumar Singh, "Quantum machine learning driven malicious user prediction for cloud network communications," *IEEE Netw. Lett.*, vol. 4, no. 4, pp. 174–178, Dec. 2022.
[172] A. Alomari and S. A. P. Kumar, "DEQSVC: Dimensionality reduction and encoding technique for quantum support vector classifier approach to detect DDoS attacks," *IEEE Access*, vol. 11, pp. 110570–110581, 2023.
[173] M. Guo, S. Pan, W. Li, F. Gao, S. Qin, X. Yu, X. Zhang, and Q. Wen, "Quantum algorithm for unsupervised anomaly detection," *Phys. A, Stat. Mech. Appl.*, vol. 625, Sep. 2023, Art. no. 129018.
[174] A. Kukliansky, M. Orescanin, C. Bollmann, and T. Huffmire, "Network anomaly detection using quantum neural networks on noisy quantum computers," *IEEE Trans. Quantum Eng.*, vol. 5, pp. 1–11, 2024.
[175] Y. Peng, X. Li, Z. Liang, and Y. Wang, "HyQ2: A hybrid quantum neural network for NextG vulnerability detection," *IEEE Trans. Quantum Eng.*, vol. 5, pp. 1–19, 2024.
[176] T. Q. Duong, J. A. Ansere, B. Narottama, V. Sharma, O. A. Dobre, and H. Shin, "Quantum-inspired machine learning for 6G: Fundamentals, security, resource allocations, challenges, and future research directions," *IEEE Open J. Veh. Technol.*, vol. 3, pp. 375–387, 2022.
[177] J. Zhang, G. Zheng, T. Koike-Akino, K.-K. Wong, and F. A. Burton, "Hybrid quantum-classical neural networks for downlink beamforming optimization," *IEEE Trans. Wireless Commun.*, vol. 23, no. 11, pp. 16498–16512, Nov. 2024.
[178] T. Liu, B. Song, F. Meng, W. Yang, J. Li, J. You, and W. Lu, "Rapid estimation method for coupling degree of airborne antenna based on quantum neural network," *IEEE Antennas Wireless Propag. Lett.*, vol. 23, no. 12, pp. 4089–4093, Dec. 2024.
[179] W. Wang, J. Xu, X. Ding, Z. Song, Y. Huang, X. Zhou, and Z. Shan, "HQNN-SFOP: Hybrid quantum neural networks with signal feature overlay projection for drone detection using radar return signals—A simulation," *Comput., Mater. Continua*, vol. 81, no. 1, pp. 1363–1390, 2024.
[180] M. Kumar, U. Dohare, S. Kumar, and N. Kumar, "Blockchain based optimized energy trading for E-mobility using quantum reinforcement learning," *IEEE Trans. Veh. Technol.*, vol. 72, no. 4, pp. 5167–5180, Apr. 2023.
[181] A. Mukherjee, V. Hassija, and V. Chamola, "QuARCS: Quantum anomaly recognition and caption scoring framework for surveillance videos," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 5618–5627, Aug. 2024.
[182] J. Amin, M. A. Anjum, K. Ibrar, M. Sharif, S. Kadry, and R. G. Crespo, "Detection of anomaly in surveillance videos using quantum convolutional neural networks," *Image Vis. Comput.*, vol. 135, Jul. 2023, Art. no. 104710.
[183] S. Park, H. Baek, J. W. Yoon, Y. K. Lee, and J. Kim, "Aqua: Analytics-driven quantum neural network (QNN) user assistance for software validation," *Future Gener. Comput. Syst.*, vol. 159, pp. 545–556, Oct. 2024.
[184] B. Narottama, T. Jamaluddin, and S. Y. Shin, "Quantum neural network with parallel training for wireless resource optimization," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 5835–5847, May 2024.
[185] M. B. A. Dastagir and D. Han, "Towards hybrid quantum-classical deep learning architecture for indoor-outdoor detection using QCNN-LSTM and cluster state signal processing," *IEEE Signal Process. Lett.*, vol. 31, pp. 2945–2949, 2024.
[186] F. Fan, Y. Shi, T. Guggemos, and X. X. Zhu, "Hybrid quantum-classical convolutional neural network model for image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 18145–18159, Dec. 2024.
[187] A. Sebastianelli, D. A. Zaidenberg, D. Spiller, B. Le Saux, and S. Ullo, "On circuit-based hybrid quantum neural networks for remote sensing imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 565–580, 2022.
[188] E. Pasetto, M. Riedel, K. Michielsen, and G. Cavallaro, "Kernel approximation on a quantum annealer for remote sensing regression tasks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3262–3269, 2024.
[189] S. Otgonbaatar, G. Schwarz, M. Datcu, and D. Kranzlmüller, "Quantum transfer learning for real-world, small, and high-dimensional remotely sensed datasets," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9223–9230, 2023.
[190] S. Otgonbaatar and M. Datcu, "Classification of remote sensing images with parameterized quantum gates," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[191] A. Miroszewski, J. Mielczarek, G. Czelusta, F. Szczepanek, B. Grabowski, B. Le Saux, and J. Nalepa, "Detecting clouds in multispectral satellite images using quantum-kernel support vector machines," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 7601–7613, 2023.
[192] F. Fan, Y. Shi, and X. X. Zhu, "Land cover classification from Sentinel-2 images with quantum-classical convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 12477–12489, 2024.
[193] S. Otgonbaatar and M. Datcu, "A quantum annealer for subset feature selection and the classification of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7057–7065, 2021.
[194] C.-H. Lin, T.-H. Lin, and J. Chanussot, "Quantum information-empowered graph neural network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537615.

[195] G. S. Kim, J. Chung, and S. Park, "Realizing stabilized landing for computation-limited reusable rockets: A quantum reinforcement learning approach," *IEEE Trans. Veh. Technol.*, vol. 73, no. 8, pp. 12252–12257, Aug. 2024.

[196] R. Bhavsar, N. K. Jadav, U. Bodkhe, R. Gupta, S. Tanwar, G. Sharma, P. N. Bokoro, and R. Sharma, "Classification of potentially hazardous asteroids using supervised quantum machine learning," *IEEE Access*, vol. 11, pp. 75829–75848, 2023.

[197] D. Lakshmi, N. Nagpal, S. Chandrasekaran, and D. J. Hemanth, "A quantum-based approach for offensive security against cyber attacks in electrical infrastructure," *Appl. Soft Comput.*, vol. 136, Mar. 2023, Art. no. 110071.

[198] M. M. Sushmit and I. M. Mahbubul, "Forecasting solar irradiance with hybrid classical–quantum models: A comprehensive evaluation of deep learning and quantum-enhanced techniques," *Energy Convers. Manage.*, vol. 294, Oct. 2023, Art. no. 117555.

[199] S. DiAdamo, C. O'Meara, G. Cortiana, and J. Bernabe-Moreno, "Practical quantum K-means clustering: Performance analysis and applications in energy grid classification," *IEEE Trans. Quantum Eng.*, vol. 3, pp. 1–16, 2022.

[200] Y. Li, R.-G. Zhou, R. Xu, J. Luo, and S.-X. Jiang, "A quantum mechanics-based framework for EEG signal feature extraction and classification," *IEEE Trans. Emerg. Topics Comput.*, vol. 10, no. 1, pp. 211–222, Jan. 2022.

[201] H. Liu, T. Yuan, X. Zhang, and H. Xu, "Quantum entanglement and self-attention neural networks: An investigation into passengers and stops characteristics for optimal bus stop localization," *Inf. Fusion*, vol. 112, Dec. 2024, Art. no. 102527.

[202] R. Harikrishnakumar and S. Nannapaneni, "Forecasting bike sharing demand using quantum Bayesian network," *Expert Syst. Appl.*, vol. 221, Jul. 2023, Art. no. 119749.

[203] C. Wang, Y. Shi, and S. Wang, "A reposition algorithm for E-hailing based on quantum annealing and intuitive reasoning," *Intell. Converged Netw.*, vol. 5, no. 4, pp. 317–335, Dec. 2024.

[204] Z. Qu, X. Liu, and M. Zheng, "Temporal–spatial quantum graph convolutional neural network based on Schrödinger approach for traffic congestion prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 8, pp. 8677–8686, Aug. 2023.

[205] E. Paquet and F. Soleymani, "QuantumLeap: Hybrid quantum neural network for financial predictions," *Expert Syst. Appl.*, vol. 195, Jun. 2022, Art. no. 116583.

[206] M. A. Jahin, M. S. H. Shovon, M. S. Islam, J. Shin, M. F. Mridha, and Y. Okuyama, "QAmplifyNet: Pushing the boundaries of supply chain backorder prediction using interpretable hybrid quantum-classical neural network," *Sci. Rep.*, vol. 13, no. 1, p. 18246, Oct. 2023.

[207] M. Akrom, S. Rustad, T. Sutojo, D. R. I. M. Setiadi, H. K. Dipojono, R. Maezono, and M. Solomon, "Quantum machine learning for corrosion resistance in stainless steel," *Mater. Today Quantum*, vol. 3, Sep. 2024, Art. no. 100013.

[208] E. Choi, J. Sul, J. E. Kim, S. Hong, B. I. Gonzalez, P. Cembellin, and Y. Wang, "Quantum machine learning for additive manufacturing process monitoring," *Manuf. Lett.*, vol. 41, pp. 1415–1422, Oct. 2024.

[209] S. C. Pravin, G. Rohith, V. Kiruthika, E. Manikandan, S. Methelesh, and A. Manoj, "Underwater animal identification and classification using a hybrid classical-quantum algorithm," *IEEE Access*, vol. 11, pp. 141902–141914, 2023.

[210] J. Kim, J. Huh, and D. K. Park, "Classical-to-quantum convolutional neural network transfer learning," *Neurocomputing*, vol. 555, Oct. 2023, Art. no. 126643.

[211] M. Junayed Hasan and M. R. C. Mahdy, "Bridging classical and quantum machine learning: Knowledge transfer from classical to quantum neural networks using knowledge distillation," 2023, *arXiv:2311.13810*.

[212] M. Alam, S. Kundu, and S. Ghosh, "Knowledge distillation in quantum neural network using approximate synthesis," 2022, *arXiv:2207.01801*.

[213] J. Wu, T. Hu, and Q. Li, "Distributed quantum machine learning: Federated and model-parallel approaches," *IEEE Internet Comput.*, vol. 28, no. 2, pp. 65–72, Mar. 2024.

[214] M. Schuld and N. Killoran, "Is quantum advantage the right goal for quantum machine learning?" *PRX Quantum*, vol. 3, no. 3, Jul. 2022, Art. no. 030101.

[215] N. M. Intelligence, "Seeking a quantum advantage for machine learning," *Nature Mach. Intell.*, vol. 5, no. 8, p. 813, Aug. 2023.

[216] *ISO/IEC TR 5469:2024 Artificial Intelligence—Functional Safety and AI Systems*, ISO/IEC Standard TR 5469:2024, 2024.

[217] *ISO/IEC 23837-1:2023 Information Security - Security Requirements, Test and Evaluation Methods for Quantum Key Distribution*, ISO/IEC Standard 23837-1:2023, 2023.

[218] J. Heredge, N. Kumar, D. Herman, S. Chakrabarti, R. Yalovetzky, S. H. Sureshbabu, C. Li, and M. Pistoia, "Characterizing privacy in quantum machine learning," *NPJ Quantum Inf.*, vol. 11, no. 1, p. 80, May 2025.

[219] M. Kop, M. Aboy, and T. Minssen, "Intellectual property in quantum computing and market power: A theoretical discussion and empirical analysis," *J. Intellectual Property Law Pract.*, vol. 17, no. 8, pp. 613–628, Jul. 2022.

[220] P. E. Vermaas, "The societal impact of the emerging quantum technologies: A renewed urgency to make quantum theory understandable," *Ethics Inf. Technol.*, vol. 19, no. 4, pp. 241–246, Dec. 2017.

[221] S. Mahajan, "The democratization dilemma: When everyone is an expert, who do we trust?" *Humanities Social Sci. Commun.*, vol. 12, no. 1, p. 455, Mar. 2025.

[222] Z. C. Seskir, S. Umbrello, C. Coenen, and P. E. Vermaas, "Democratization of quantum technologies," *Quantum Sci. Technol.*, vol. 8, no. 2, Feb. 2023, Art. no. 024005.

[223] E. Perrier, "The quantum governance stack: Models of governance for quantum information technologies," *Digit. Soc.*, vol. 1, no. 3, p. 22, Oct. 2022.

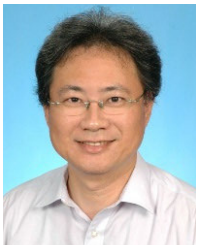
[224] M. Kop, "Quantum-ELSPI: A novel field of research," *Digit. Soc.*, vol. 2, no. 2, p. 20, May 2023.

[225] M. Kop, "Ethics in the quantum age," *Phys. World*, vol. 34, no. 12, p. 31, Dec. 2021.

[226] B. Mehta, B. Tuscai, T. R. Wang, and K. Mahady, "Responsible quantum readiness: A perspective from financial services organization," in *Proc. IEEE Int. Conf. Quantum Comput. Eng. (QCE)*, Sep. 2024, pp. 71–76.

[227] E. Nabavi and C. Browne, "Leverage zones in responsible AI: Towards a systems thinking conceptualization," *Humanities Social Sci. Commun.*, vol. 10, no. 1, p. 82, Mar. 2023.

[228] L. M. Possati, "Ethics of quantum computing: An outline," *Philosophy Technol.*, vol. 36, no. 3, p. 48, Jul. 2023.



YING-YI HONG (Senior Member, IEEE) received the B.S.E.E. degree from Chung Yuan Christian University (CYCU), Taiwan, in 1984, the M.S.E.E. degree from National Cheng Kung University (NCKU), Taiwan, in 1986, and the Ph.D. degree from the Department of Electrical Engineering, National Tsing Hua University (NTHU), Taiwan, in December 1990. Sponsored by the Ministry of Education, Taiwan, he conducted research with the Department of Electrical Engineering, University of Washington, Seattle, WA, USA, from August 1989 to August 1990. He has been with CYCU, since 1991, where he was the Dean of the College of Electrical Engineering and Computer Science, from 2006 to 2012. He was promoted to the Chair Professor due to his exceptional research, leadership, teamwork, and international collaboration, in 2024. From 2012 to 2018, he was a General Secretary with CYCU, where he is currently the Vice President. His research interests include power system analysis and artificial intelligence applications. He received the Outstanding Professor of the Electrical Engineering Award from Chinese Institute of Electrical Engineering (CIEE), Taiwan, in 2006, and the Outstanding Professor of the Engineering Award from Chinese Institute of Engineers (CIE), Taiwan, in 2024. He was the Chair of the IEEE PES Taipei Chapter, in 2001.

DYLAN JOSH DOMINGO LOPEZ (Member, IEEE) received the B.S. degree in computer engineering from Adamson University, Philippines, in 2018, the M.Sc. degree in computer science and information engineering from Chung Hua University, Taiwan, in 2020. He is currently pursuing the Ph.D. degree in EE with Chung Yuan Christian University (CYCU). He is a part-time Instructor with De La Salle University, Manila, Philippines. His research interests include optimization, quantum computing, and artificial intelligence applications.